

# **MALWARE ANALYSIS AND REVERSE ENGINEERING**

## **Snake Keylogger: Complete Static, Dynamic Analysis**

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Environment: REMnux + Windows 10 VM (VirtualBox)

Date: April 2026

**Malware Family:** Snake Keylogger

**SHA-256:**

4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083  
d154310f272e4

## **EXECUTIVE SUMMARY**

This report documents a complete malware analysis laboratory exercise performed on a Snake Keylogger sample sourced from MalwareBazaar. The analysis was conducted in an isolated virtual environment using REMnux (Linux-based analysis platform) and Windows 10 as the victim machine. The investigation covered three analytical phases: static analysis, dynamic analysis, and hybrid analysis using ANY.RUN sandbox.

The malware sample was confirmed to be Snake Keylogger — a well-known information-stealing trojan that captures keystrokes, exfiltrates credentials, and communicates with Command & Control (C2) infrastructure. The malware is compiled using AutoIt3 scripting language and employs process hollowing through RegSvcs.exe for evasion.

## **KEY FINDINGS AT A GLANCE**

File: PE32 executable | Language: AutoIt3 v3.3.3.16 | Compile Date: Feb 25, 2026 Behavior: Keylogging, Registry persistence, Process injection, C2 communication Persistence: Startup folder via unbarricading.vbs Evasion: Process hollowing via RegSvcs.exe

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# 1. OBJECTIVES & LEARNING OUTCOMES

The objective of this laboratory exercise was to execute and analyze a real-world malware sample in a controlled and isolated virtual environment using industry-standard tools. The analysis focused on three core phases — static, dynamic, and hybrid — culminating in a comprehensive report comparing findings against published threat intelligence.

## Learning Outcomes

- Set up a safe virtual machine environment for malware testing using REMnux and Windows 10
- Perform static analysis by examining the file without executing it
- Perform dynamic analysis by monitoring behavior after execution in an isolated environment
- Monitor file system, registry, process, and network changes in real time
- Collect forensic evidence including memory artifacts and system logs
- Combine static and dynamic findings to understand the full malware behavior chain
- Write a structured, professional malware analysis report
- Design a comprehensive malware incident response plan
- 

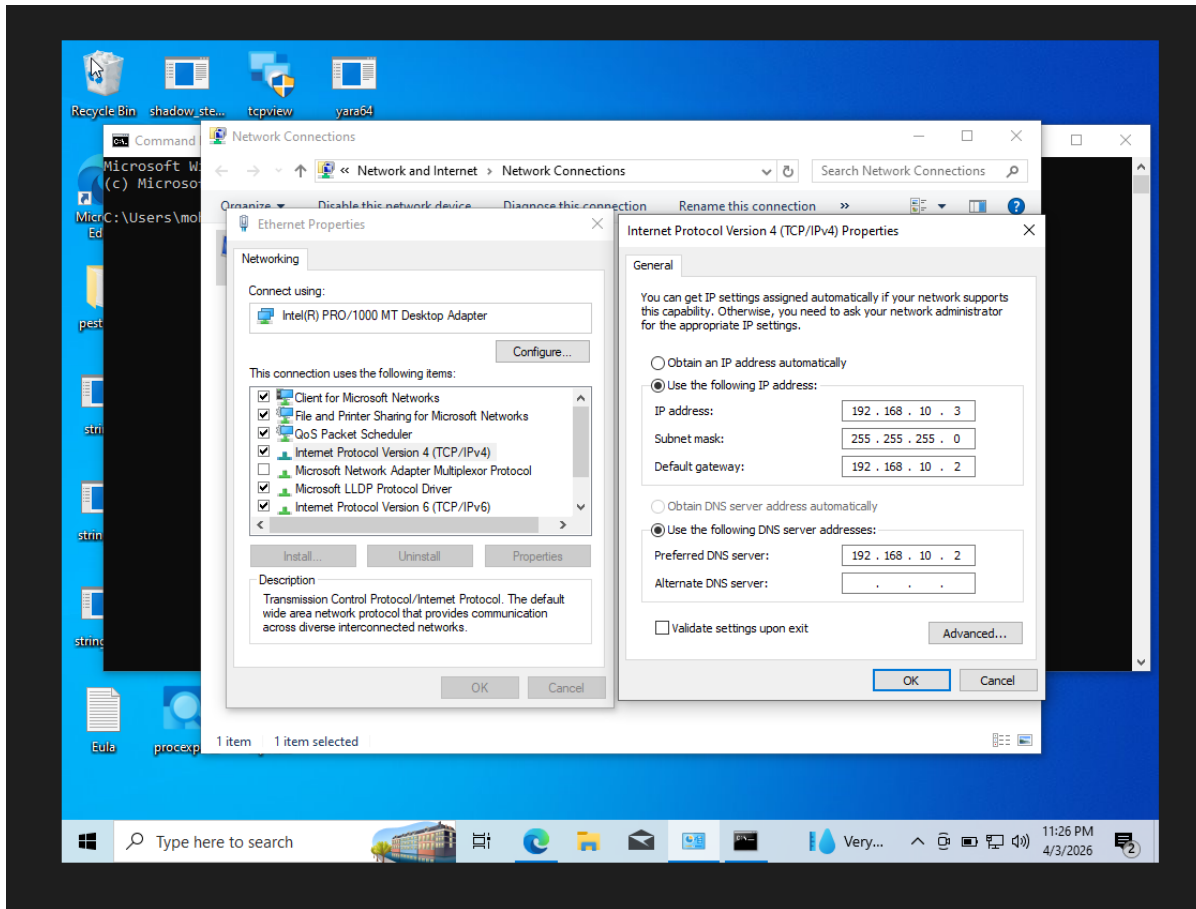
## 2. ENVIRONMENT SETUP

A safe sandbox environment was constructed using two isolated virtual machines: REMnux (a Linux-based malware analysis toolkit) and Windows 10 (the victim machine for malware execution). Both machines were connected via a Host-Only network adapter, ensuring the malware could not reach the real internet or any live systems.

### 2.1 REMnux Virtual Machine

REMnux is a Linux-based distribution specifically designed for malware analysis. It provides pre-installed tools for reverse engineering, network simulation, and behavioral analysis. The VM was downloaded from [remnux.org](https://remnux.org) as a VirtualBox OVA appliance and imported with the following configuration:

| Setting             | Value                          |
|---------------------|--------------------------------|
| RAM                 | 4096 MB                        |
| CPU                 | 2 Cores                        |
| Network Adapter     | Host-Only Adapter (VirtualBox) |
| IP Address (Static) | 192.168.10.2/24                |
| Gateway             | 192.168.10.1                   |
| DNS Server          | 192.168.10.1 (self)            |
| OS                  | Ubuntu 24.04 (Noble Numbat)    |



```
C:\Users\mohamed>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:
Reply from 192.168.10.2: bytes=32 time=1ms TTL=64
Reply from 192.168.10.2: bytes=32 time=1ms TTL=64
Reply from 192.168.10.2: bytes=32 time=4ms TTL=64
Reply from 192.168.10.2: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 1ms

C:\Users\mohamed>
```

A static IP was assigned to REMnux so the Windows victim machine can always communicate with it reliably. Without a static IP, the malware network simulation would fail because the DNS and HTTP fake services would be unreachable.

```
Apr 4 10:57
remnux@remnux: ~
remnux@remnux:~$ sudo ip addr add 192.168.10.2/24 dev enp0s3
remnux@remnux:~$ sudo ip link set enp0s3 up
remnux@remnux:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:e0:4f:a2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.2/24 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fee0:4fa2/64 scope link
        valid_lft forever preferred_lft forever
remnux@remnux:~$ ping 192.168.10.3
PING 192.168.10.3 (192.168.10.3) 56(84) bytes of data.
64 bytes from 192.168.10.3: icmp_seq=1 ttl=128 time=12.5 ms
64 bytes from 192.168.10.3: icmp_seq=2 ttl=128 time=1.34 ms
```

## 2.2 Windows 10 Victim Virtual Machine

A fresh Windows 10 Pro (64-bit) virtual machine was created as the victim environment where the malware would be executed. Windows Defender and Windows Firewall were disabled to allow unrestricted malware execution for analysis purposes.

| Setting             | Value                          |
|---------------------|--------------------------------|
| VM Name             | Windows 10 Victim              |
| ISO File            | Windows 10 Pro x64 (Dec 2023)  |
| RAM                 | 2048 MB                        |
| Hard Disk           | 50 GB                          |
| CPU                 | 2 Cores                        |
| Network Adapter     | Host-Only Adapter (VirtualBox) |
| IP Address (Static) | 192.168.10.3                   |
| Default Gateway     | 192.168.10.2 (REMnux)          |
| DNS Server          | 192.168.10.2 (REMnux/INetSim)  |

The Windows machine was configured to route all DNS and internet traffic through REMnux (192.168.10.2), which runs INetSim to simulate fake internet services. This allows the malware to attempt network connections while all traffic is safely captured and logged.

## 2.3 Network Configuration & INetSim

INetSim (Internet Services Simulator) was installed on REMnux to simulate common internet services — including HTTP, HTTPS, DNS, SMTP, and FTP. When the malware attempts to connect to a Command & Control (C2) server or send stolen data, all requests are captured by INetSim and logged for analysis.

### Why Host-Only Network?

A host-only network creates an isolated boundary — malware can only interact between the two VMs and cannot reach the analyst's real machine or the internet. Combined with INetSim, this allows full network activity capture without any risk of real-world damage.

## 3. MALWARE DOWNLOAD & PREPARATION

The malware sample was sourced from MalwareBazaar (bazaar.abuse.ch) — a public malware repository operated by Abuse.ch. The sample was identified as Snake Keylogger, a well-documented information-stealing trojan.

| Attribute      | Value                                                            |
|----------------|------------------------------------------------------------------|
| Source         | MalwareBazaar (bazaar.abuse.ch)                                  |
| SHA-256        | 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4 |
| MD5            | 8bc5d0081f7c99103331cb408e45601a                                 |
| SHA-1          | 796ade52d238971ed98543c7b4670f5e746d22b1                         |
| File Type      | PE32 executable (GUI) Intel 80386, MS Windows                    |
| File Size      | 1,118,720 bytes (1.07 MB)                                        |
| ZIP Password   | infected (standard MalwareBazaar password)                       |
| Compile Date   | Wed Feb 25 07:35:38 2026 UTC                                     |
| Malware Family | Snake Keylogger                                                  |
| Packer/Builder | C++ compiled binary                                              |

```
Apr 4 14:34
remnux@remnux: /media/sf_keylog

remnux@remnux:~$ cd /media
remnux@remnux:/media$ ls
remnux sf_kalishare sf_keylog
remnux@remnux:/media$ cd keylog
bash: cd: keylog: No such file or directory
remnux@remnux:/media$ cd sf_keylog
remnux@remnux:/media/sf_keylog$ ls
'System Volume Information'  unbaricading.exe
remnux@remnux:/media/sf_keylog$
```

```
remnux@remnux: ~

remnux@remnux:~$ ls
Desktop  Downloads  Pictures  Templates  keylog.exe.mal  unbaricading.zip
Documents  Music      Public    Videos    malware
remnux@remnux:~$
```

```

remnux@remnux:~/malware$ 7z x 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.zip
7-Zip 23.01 (x84) : Copyright (c) 1999-2023 Igor Pavlov : 2023-06-28
64-bit locale=C.UTF-8 Threads:2 OPEN_MAX:1024

Scanning the drive for archives:
1 file, 666331 bytes (651 KiB)

Extracting archive: 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.zip
..
Path = 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.zip
Type = zip
Physical Size = 666331

Would you like to replace the existing file:
Path:      ./4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
Size:      0 bytes
Modified:  2026-04-04 09:43:12
with the file from archive:
Path:      4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
Size:      1118720 bytes (1093 KiB)
Modified:  2026-04-04 09:43:12
? (Y)es / (N)o / (A)lways / (S)kip all / A(u)to rename all / (Q)uit? y

Enter password (will not be echoed):
Everything is Ok

Size:      1118720
Compressed: 666331

```

The sample was downloaded as a password-protected ZIP file and transferred to both VMs using VirtualBox Shared Folders. On REMnux, the following commands were used to prepare the sample:

```

# Add REMnux user to vboxsf group for shared folder access sudo adduser remnux vboxsf
# Copy malware ZIP from shared folder cp
/media/sf_MalwareShare/4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.zip ~/Downloads/keylogger.zip
# Navigate to Downloads cd ~/Downloads # Install 7-zip sudo apt install p7zip-full # Extract with password 7z x keylogger.zip
# Enter password: infected # Verify file type file 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe

```

## 4. STATIC MALWARE ANALYSIS

Static malware analysis involves examining a malware file without executing it. This approach allows analysts to safely study the file's properties, code structure, embedded strings, imported libraries, and metadata. Static analysis provides a foundational understanding of the malware's capabilities without any risk of system infection.

### 4.1 File Identification & Hash Analysis

The first step in static analysis is computing the file's cryptographic hashes. These serve as unique fingerprints that allow analysts to identify

```
remnux@remnux:~/malware$ pecheck 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
PE check for '4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe':
Entropy: 6.968161 (Min=0.0, Max=8.0)
Size: 1118720
MD5      hash: 8bc5d0081f7c99103331cb408e45601a
SHA-1    hash: 796ade52d238971ed98543c7b4670f5e746d22b1
SHA-256  hash: 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4
SHA-512  hash: 9fc6b573af5865795e66f1c8b46a46fc45b46b7837a19dea589c1f77621edbc40c1e323b55f89fb9114822cd26
140034ecf743da5063a66cc564c1208458e744
.text entropy: 6.665680 (Min=0.0, Max=8.0)
.rdata entropy: 5.692798 (Min=0.0, Max=8.0)
.data entropy: 0.584577 (Min=0.0, Max=8.0)
.rsrc entropy: 7.797005 (Min=0.0, Max=8.0)
.reloc entropy: 6.798204 (Min=0.0, Max=8.0)
```

the malware family, search for threat intelligence, and verify file integrity.

```
remnux@remnux:~/malware$ ls -l
total 1748
-rw-r--r-- 1 remnux remnux 1118720 Apr  4 09:43 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
-rwxr-x-- 1 remnux remnux 666331 Apr  4 09:51 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.zip
remnux@remnux:~/malware$

remnux@remnux:~/malware$ file 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe: PE32 executable (GUI) Intel 80386, for MS Windows, 5 sections
remnux@remnux:~/malware$
```

The hash values are used to search threat intelligence databases such as VirusTotal, AlienVault OTX, and MalwareBazaar to confirm the malware family and retrieve prior analysis reports.

## 4.2 Entropy & PE Section Analysis (pecheck)

PEcheck was used to analyze the Portable Executable (PE) structure of the malware. Entropy analysis measures the randomness of data within each section — a high entropy value (above 7.0) strongly indicates that data has been compressed or encrypted to hide its true content.

pecheck 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe

| PE Property       | Finding                                                 |
|-------------------|---------------------------------------------------------|
| File Entropy      | 6.968 / 8.0 — Likely packed / obfuscated                |
| .text Entropy     | 6.665 — Executable code section                         |
| .rdata Entropy    | 5.692 — Read-only data (strings/constants)              |
| .data Entropy     | 0.584 — Global variables                                |
| .rsrc Entropy     | 7.797 — HIGH: Possible hidden payload or encrypted data |
| .reloc Entropy    | 6.798 — Relocation data                                 |
| Magic Bytes       | 0x5A4D (MZ) — Valid Windows PE executable               |
| Architecture      | 32-bit (IMAGE_FILE_32BIT_MACHINE)                       |
| Compile Timestamp | Wed Feb 25 07:35:38 2026 UTC                            |

### HIGH ENTROPY WARNING

The .rsrc section has entropy of 7.797 out of a maximum of 8.0. This is a major red flag indicating the malware has hidden its primary payload, encrypted configuration data, or embedded malicious scripts inside the resource section. Legitimate programs rarely have resource section entropy above 7.0.

## 4.3 DLL Import Analysis

The imported DLL libraries reveal the capabilities of the malware — what Windows functions it can call and therefore what actions it can perform. Three particularly suspicious libraries were identified:

| Library      | Malicious Use                                                                        |
|--------------|--------------------------------------------------------------------------------------|
| USER32.dll   | Keyboard and mouse input capture — GetAsyncKeyState, SendInput. Used for keylogging. |
| WSOCK32.dll  | Network socket communication — send, recv, connect. Used to exfiltrate stolen data.  |
| ADVAPI32.dll | Registry manipulation — RegCreateKeyExW, RegSetValueExW. Used for persistence.       |
| KERNEL32.dll | File operations, process creation, memory management — core Windows functions.       |
| IPHLPAPI.dll | IP Helper API — used to enumerate network interfaces and gather system information.  |

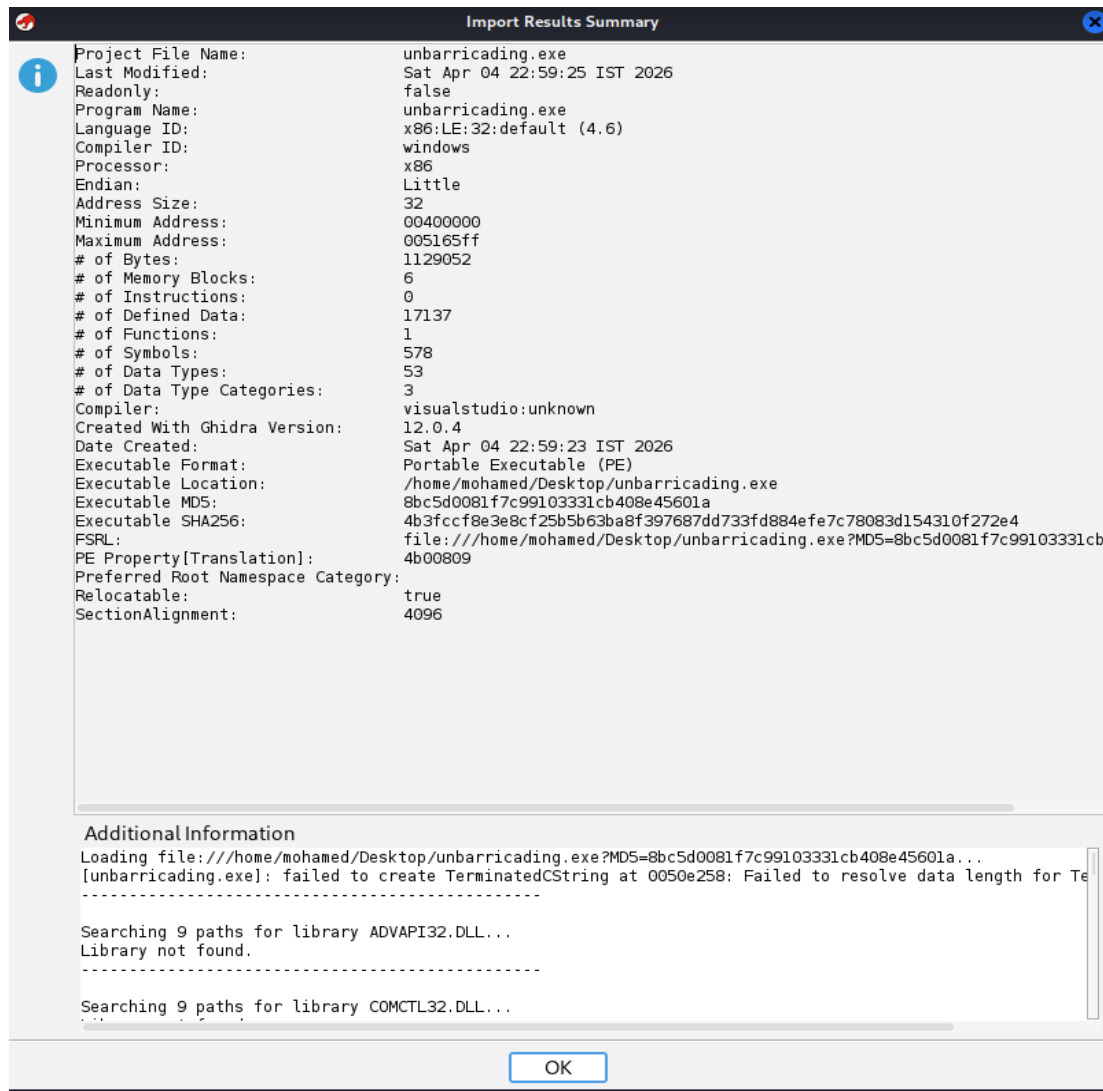
## 4.4 PEiD Detection with Exeinfo PE

Exeinfo PE was used on the Windows 10 victim VM to identify the packer or compiler used to build the malware. This information is critical because knowing the builder helps analysts select the correct unpacking or decompilation strategy.

## 4.5 Disassembly with Ghidra

Ghidra (NSA's open-source reverse engineering framework) was used to disassemble and decompile the malware binary into human-readable assembly code and pseudo-C. This allows deep analysis of the malware's internal logic, function calls, and data flow.

Setup: Ghidra v12.0.3 was installed on the Windows 10 VM along with Java OpenJDK 25. The malware was imported as a Portable Executable (PE) project.



File Edit Analysis BSIm Graph Navigation Search Select Tools Window Help

Listing: unbarricading.exe

Symbol Tree

- > f 'vector\_destructor\_iterator'
- > a
- > f BuildCatchObjectInternal<struct...
- > C
- > c
- > de
- > e
- > f ExFilterRethrow
- > f
- > F
- > g
- > f GetTableIndexFromLocaleName
- > i
- > f InternalCompareStringA
- > f LongLongSub
- > f make\_c\_string\_character\_source
- > o
- > parse\_
- > re
- > s

Filter:

Data Type Manager

Data Types

- > BuiltInTypes
- > unbarricading.exe
  - > basetsd.h
  - > commctrl.h
  - > commdlg.h
  - > crtdefs.h
  - > Demangler
  - > DOS

Filter:

```
// Headers
// ram:00400000-ram:004003ff

assume DF = 0x0 (Default)
IMAGE_DOS_HEADER_00400000.e_res[4][2]
IMAGE_DOS_HEADER_00400000.e_lfanew
IMAGE_DOS_HEADER_00400000

XREF[15,2]: 00400154(*),
FUN_0041e825:0041e952(*),
__scrt_is_nonwritable_in_current...
__scrt_is_nonwritable_in_current...
__scrt_is_nonwritable_in_current...
__scrt_common_main_seh:00420469(...
CATLBaseModule:00420cca(*),
CATLBaseModule:00420cda(*),
CATLBaseModule:00420cda(*),
__IsNonwritableInCurrentImage:00...
__IsNonwritableInCurrentImage:00...
__IsNonwritableInCurrentImage:00...
FUN_004099d0:00469980(*),
FUN_0041b728:00458920(*),
FUN_0041b728:004589f5(*),
__scrt_is_nonwritable_in_current...
FUN_0047902a:00479390(*)

00400000 4d 5a 90 IMAGE_DOS...
00 03 00
00 00 04 ...

00400000 4d 5a char[2] "MZ" e_magic XREF[15]: 00400154(*),
FUN_0041e825:0041e952(*),
__scrt_is_nonwritab
__scrt_is_nonwritab
__scrt_is_nonwritab
__scrt_common_main_s
CATLBaseModule:00420
CATLBaseModule:00420
CATLBaseModule:00420
__IsNonwritableInCur
__IsNonwritableInCur
__IsNonwritableInCur
FUN_004099d0:0046998
FUN_0041b728:0045892
FUN_0041b728:004589f
FUN_00400154(*),
FUN_0041e825:0041e
__scrt_is_nonwri
__scrt_is_nonwri
__scrt_is_nonwri
```

00400000 4d char 'M' [0] XREF[15]: 00400154(\*),
FUN\_0041e825:0041e
\_\_scrt\_is\_nonwri
\_\_scrt\_is\_nonwri
\_\_scrt\_is\_nonwri

Defined Strings - 3 items (of 2121)

| Location | String Value     | String Representation | Data Type |
|----------|------------------|-----------------------|-----------|
| 004c2bf8 | APPDATACOMMONDIR | u"APPDATACOMMONDIR"   | unicode   |
| 004c2d00 | LOCALAPPDATA\DIR | u"LOCALAPPDATA\DIR"   | unicode   |
| 004c2d20 | APPDATA\DIR      | u"APPDATA\DIR"        | unicode   |

Filter: appdata

Decompile: append - (unbarricading.exe)

CodeBrowser: keylog/unbarricading.exe

File Edit Analysis BSim Graph Navigation Search Select Tools Window Help

Symbol Tree

- Functions
  - vector\_destructor\_iterator
  - a
  - append
  - ateviv\_exception\_filter
  - BuildCatchObjectInternal<struct...
  - param\_1
  - param\_2
  - param\_3
  - param\_4
  - local\_8
  - local\_14
  - CallCatchBlock
  - Catch\_Ali@004243b1
  - CatchIt<class \_\_FrameHandle
  - CPItoLocaleName
  - c
  - common\_atof\_1<>

Filter:

Data Type Manager

Data Types

- BuiltInTypes
- unbarricading.exe
- basetsd.h
- commctrl.h
- commdlg.h
- crtdfs.h
- Demangler
- none

Filter:

Listing: unbarricading.exe

| Address           | Disassembly                   | Comment | XREF     | Value       |
|-------------------|-------------------------------|---------|----------|-------------|
| 004c2c8f 00       | ?? 00h                        |         |          |             |
| 004c2c90 50 00 52 | u_PROGRAMSCOMMONDIR_004c2c90  |         | XREF[1]: | 004b1a58(*) |
| 004c2c90 00 4f 00 | unicode u"PROGRAMSCOMMONDIR"  |         |          |             |
| 004c2c90 47 00 52 | ?? 00h                        |         |          |             |
| 004c2c94 53 00 54 | u_STARTMENUCOMMONDIR_004c2c94 |         | XREF[1]: | 004b1a5c(*) |
| 004c2c94 00 4f 00 | unicode u"STARTMENUCOMMONDIR" |         |          |             |
| 004c2c94 53 00 54 | ?? 00h                        |         |          |             |
| 004c2c94 00 4f 00 | ?? 00h                        |         |          |             |
| 004c2c9c 53 00 54 | u_STARTUPCOMMONDIR_004c2c9c   |         | XREF[1]: | 004b1a70(*) |
| 004c2c9c 00 4f 00 | unicode u"STARTUPCOMMONDIR"   |         |          |             |
| 004c2c9c 53 00 54 | ?? 00h                        |         |          |             |
| 004c2c9c 00 4f 00 | ?? 00h                        |         |          |             |
| 004c2c9c 4c 00 4f | u_LOCALAPPDATAIDIR_004c2d00   |         | XREF[1]: | 004b1a74(*) |
| 004c2c9c 00 43 00 | unicode u"LOCALAPPDATAIDIR"   |         |          |             |
| 004c2c9c 41 00 4c | ?? 00h                        |         |          |             |
| 004c2d00 41 00 50 | u_APPDATAIDIR_004c2d00        |         | XREF[1]: | 004b1a78(*) |
| 004c2d00 00 50 00 | unicode u"APPDATAIDIR"        |         |          |             |
| 004c2d00 44 00 41 | ?? 00h                        |         |          |             |
| 004c2d00 00 50 00 | ?? 00h                        |         |          |             |
| 004c2d00 44 00 45 | u_DESKTOPDIR_004c2d38         |         | XREF[1]: | 004b1a7c(*) |
| 004c2d00 00 53 00 | unicode u"DESKTOPDIR"         |         |          |             |
| 004c2d00 4b 00 54 | ?? 00h                        |         |          |             |
| 004c2d00 00 53 00 | ?? 00h                        |         |          |             |
| 004c2d00 4b 00 54 | ?? 00h                        |         |          |             |
| 004c2d00 46 00 41 | u_FAVORITESIDIR_004c2d50      |         | XREF[1]: | 004b1a80(*) |
| 004c2d00 00 56 00 | unicode u"FAVORITESIDIR"      |         |          |             |
| 004c2d00 4f 00 52 | ?? 00h                        |         |          |             |
| 004c2d00 00 56 00 | ?? 00h                        |         |          |             |
| 004c2d00 4f 00 52 | ?? 00h                        |         |          |             |

Defined Strings - 3 items (of 212)

| Loc...   | String Val... | String Re...  | Data Type |
|----------|---------------|---------------|-----------|
| 004c2bfb | APPDATA...    | u"APPDATA..." | unicode   |
| 004c2d00 | LOCALAP...    | u"LOCALA..."  | unicode   |
| 004c2d20 | APPDATA...    | u"APPDATA..." | unicode   |

Filter: appdata

Decompile: appen...

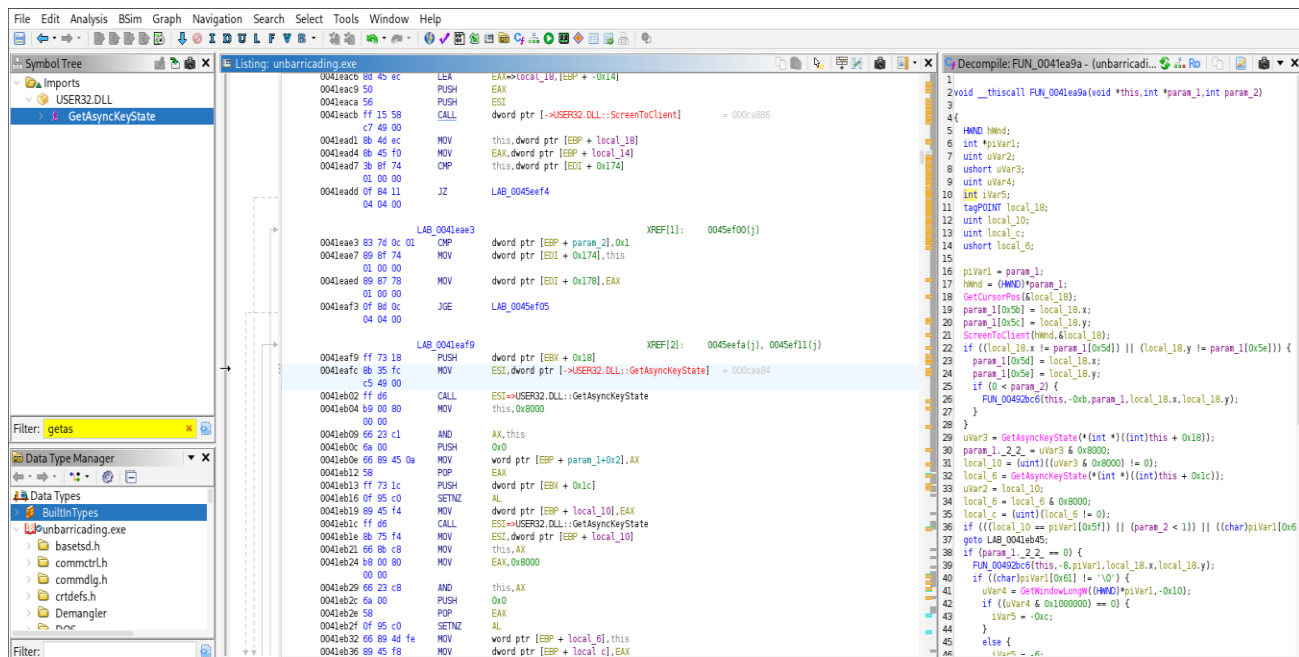
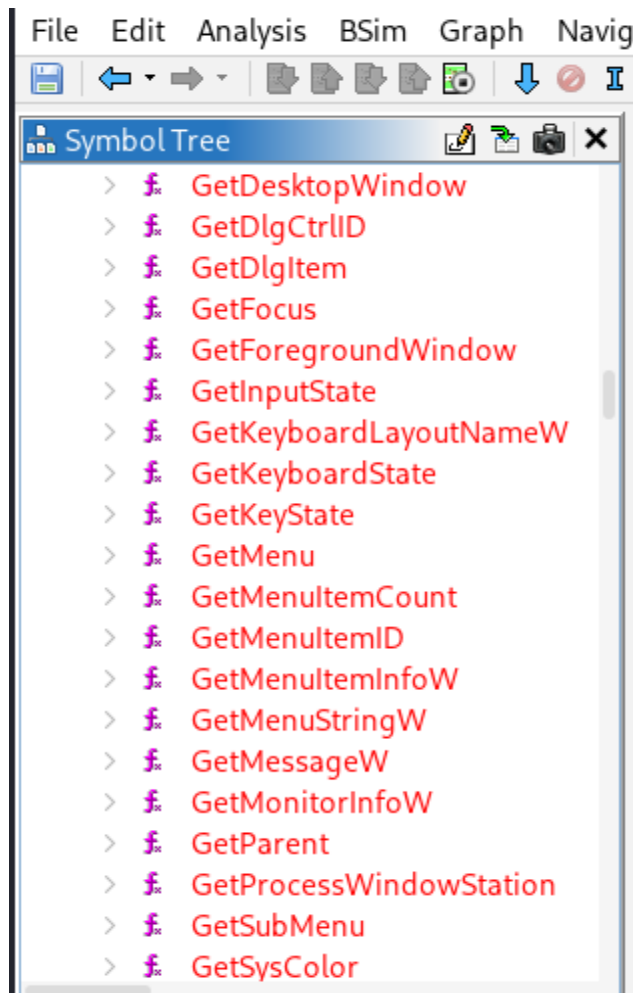
```
21
22 /* Library Function - Multiple Matches With Same Name */
23 public: int __thiscall "anonymous namespace".
24 public: int __thiscall "anonymous namespace".
25 public: int __thiscall "anonymous namespace".
26 public: int __thiscall "anonymous namespace".
27
28 Library: Visual Studio 2015 Release */
29
30 __thiscall append(void *this, void *param_1)
31
32 int iVar1;
33
34 if (iVar1 == 0) {
35     iVar1 = expand_if_necessary(this);
36     if (iVar1 == 0) {
37         *(Undefined4 *)((int)this + 4) = param_1;
38         *(int *)((int)this + 4) = *(int *)((int)this +
39         iVar1 = 0;
40     }
41     else {
42         param_1 = expand_if_necessary(this);
43     }
44 }
```

References to u\_LOCALAPPDATAIDIR\_004c2d00 - 1 locations [CodeBrowser: keylog/unbarricading.exe]

Edit Help

References to u\_LOCALAPPDATAIDIR\_004c2d00 - 1 locations

| Location | Label                            | Code Unit | Context |
|----------|----------------------------------|-----------|---------|
| 004b1a74 | addr u_LOCALAPPDATAIDIR_004c2d00 | DATA      |         |



File Edit Analysis BSIm Graph Navigation Search Select Tools Window Help

SymbolTree Listing: unbarricading.exe

Filter: fop

Data Type Manager

Data Types

BuiltInTypes

unbarricading.exe

basetsd.h

commctrl.h

commdlg.h

crtdefs.h

Demangler

Filter:

Decompile: FUN\_0040686d - (unbarricading.exe)

```

1 int __thiscall FUN_0040686d(void *this, wchar_t *param_1, int param_2)
2 {
3     FILE *pVar1;
4     MODULE pHVar2;
5     int iVar3;
6     undefined1 local_18 [16];
7     undefined4 local_8;
8     FUN_00406832(&local_8);
9     pVar1 = _wopen(param_1, L"rb");
10    *(&FILE **)(int)this + 0xc = pVar1;
11    if (pVar1 == (FILE *)0x0) {
12        FUN_004068db(this);
13        FUN_004067fb(&local_8);
14        iVar3 = 1;
15    }
16    else {
17        pHVar2 = LoadLibraryEtw(param_1, (HANDLE)0x0, 2);
18        *(&MODULE **)(int)this + 0xc = pHVar2;
19        FUN_004067fb(&local_8);
20        iVar3 = FUN_00406920(this, param_2 == 1);
21        if (iVar3 == 0) {
22            FUN_00406a95(this, local_18, 1, 0x10, *(FILE **)(int)this);
23            FUN_0047281c((int)local_18, 0x10, 0x99f2);
24            *(&undefined4 **)(int)this + 8 = 0;
25            iVar3 = FUN_00406a93((int)this);
26            *(&int **)(int)this + 4 = iVar3;
27            iVar3 = 0;
28        }
29        else {
30            FUN_004068db(this);
31        }
32    }
33    return iVar3;
34 }
35
36
37
38

```

00406887 FUN\_0040686d CALL 0x0042e57b

CodeBrowser: keylog/unbarricading.exe

File Edit Analysis BSIm Graph Navigation Search Select Tools Window Help

SymbolTree Listing: unbarricading.exe

Filter: fop

Data Type Manager

Data Types

BuiltInTypes

unbarricading.exe

basetsd.h

commctrl.h

commdlg.h

crtdefs.h

Demangler

Filter:

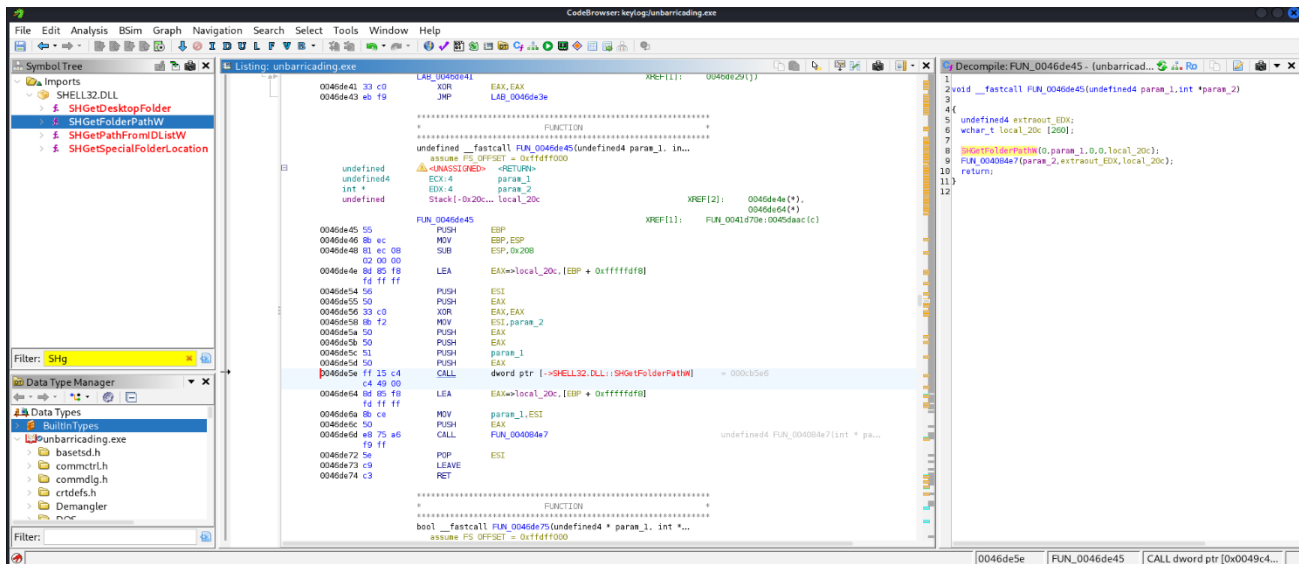
Decompile: FUN\_0040686d - (unbarricading.exe)

```

1 int __thiscall FUN_0040686d(void *this, wchar_t *param_1, int param_2)
2 {
3     FILE *pVar1;
4     MODULE pHVar2;
5     int iVar3;
6     undefined1 local_18 [16];
7     undefined4 local_8;
8     FUN_00406832(&local_8);
9     pVar1 = _wopen(param_1, L"rb");
10    *(&FILE **)(int)this + 0xc = pVar1;
11    if (pVar1 == (FILE *)0x0) {
12        FUN_004068db(this);
13        FUN_004067fb(&local_8);
14        iVar3 = 1;
15    }
16    else {
17        pHVar2 = LoadLibraryEtw(param_1, (HANDLE)0x0, 2);
18        *(&MODULE **)(int)this + 0xc = pHVar2;
19        FUN_004067fb(&local_8);
20        iVar3 = FUN_00406920(this, param_2 == 1);
21        if (iVar3 == 0) {
22            FUN_00406a95(this, local_18, 1, 0x10, *(FILE **)(int)this);
23            FUN_0047281c((int)local_18, 0x10, 0x99f2);
24            *(&undefined4 **)(int)this + 8 = 0;
25            iVar3 = FUN_00406a93((int)this);
26            *(&int **)(int)this + 4 = iVar3;
27            iVar3 = 0;
28        }
29        else {
30            FUN_004068db(this);
31        }
32    }
33    return iVar3;
34 }
35
36
37
38

```

0040686d FUN\_0040686d PUSH EBP



## Program Tree — PE Sections

The Program Tree in Ghidra provides a structural view of all PE sections. Each section serves a specific role, and understanding them helps identify where malicious code and data are hidden:

| Section    | Malicious Use                                                           |
|------------|-------------------------------------------------------------------------|
| Headers    | Basic file metadata to make malware appear as a normal program          |
| .text      | Primary executable code — performs harmful actions (main malware logic) |
| .rdata     | Hidden strings including URLs, C2 addresses, commands                   |
| .data      | Variables used to store stolen data during program execution            |
| .rsrc      | Hidden payload, encrypted files, embedded AutoIt scripts (HIGH entropy) |
| .reloc     | Relocation data for memory loading                                      |
| Debug Data | Development artifacts (may contain leaked information)                  |
| .tls       | Thread Local Storage — can execute code before main() for anti-analysis |

## Symbol Tree — Imported Functions (ADVAPI32.DLL)

ADVAPI32.DLL is a critical red flag in malware analysis:

- RegCreateKeyExW — Creates new registry key for malware persistence (auto-start on boot)
- RegSetValueExW — Sets registry value to point to malware executable
- RegDeleteKeyW — Deletes registry keys to hide evidence
- AdjustTokenPrivileges — Escalates privileges to access protected system areas
- CreateProcessAsUserW — Creates processes with elevated privileges

USER32.DLL function analysis revealed:

- GetAsyncKeyState — Called 263 times across 7 functions; core keylogging mechanism
- FindWindowW / FindWindowExW — Scans for active windows to track user activity
- GetActiveWindow — Identifies the currently focused application
- BlockInput — Can block keyboard/mouse input to prevent user interference

## Keylogger Core Function (FUN\_00469b97)

The decompiled function FUN\_00469b97 is the primary keylogging routine. It uses GetAsyncKeyState to poll the state of every key (0x01 to 0xFF) in a continuous loop. When a key press is detected (bit flag 0x8000 set), the character is captured and appended to an in-memory buffer. Special handling for the Ctrl key (0x11) and modifier keys indicates the malware captures full keyboard shortcuts in addition to individual keystrokes.

# Function Graph

```
Command Prompt
C:\Users\mohamed>netstat -ano

Active Connections

Proto Local Address           Foreign Address         State       PID
TCP   0.0.0.0:135              0.0.0.0:0              LISTENING   964
TCP   0.0.0.0:445              0.0.0.0:0              LISTENING    4
TCP   0.0.0.0:5040             0.0.0.0:0              LISTENING  5056
TCP   0.0.0.0:49664            0.0.0.0:0              LISTENING   736
TCP   0.0.0.0:49665            0.0.0.0:0              LISTENING   564
TCP   0.0.0.0:49666            0.0.0.0:0              LISTENING  1232
TCP   0.0.0.0:49667            0.0.0.0:0              LISTENING  1292
TCP   0.0.0.0:49668            0.0.0.0:0              LISTENING  2672
TCP   0.0.0.0:49669            0.0.0.0:0              LISTENING   712
TCP   0.0.0.0:64230            0.0.0.0:0              LISTENING  1732
TCP   192.168.10.3:139         0.0.0.0:0              LISTENING    4
TCP   [::]:135                 [::]:0                 LISTENING   964
TCP   [::]:445                 [::]:0                 LISTENING    4
TCP   [::]:49664               [::]:0                 LISTENING   736
TCP   [::]:49665               [::]:0                 LISTENING   564
TCP   [::]:49666               [::]:0                 LISTENING  1232
TCP   [::]:49667               [::]:0                 LISTENING  1292
TCP   [::]:49668               [::]:0                 LISTENING  2672
TCP   [::]:49669               [::]:0                 LISTENING   712
TCP   [::]:64230               [::]:0                 LISTENING  1732
UDP   0.0.0.0:123              *:                      8084
UDP   0.0.0.0:500              *:                      4544
UDP   0.0.0.0:4500             *:                      4544
UDP   0.0.0.0:5050             *:                      5056
UDP   0.0.0.0:5353             *:                      2344
UDP   0.0.0.0:5355             *:                      2344
UDP   0.0.0.0:52024            *:                      1576
UDP   0.0.0.0:54400            *:                      1576
UDP   0.0.0.0:54766            *:                      1576
UDP   0.0.0.0:55774            *:                      1576
UDP   127.0.0.1:1900           *:                      3332
UDP   127.0.0.1:55046          *:                      3332
UDP   127.0.0.1:56693          *:                      1516
UDP   192.168.10.3:137         *:                      4
UDP   192.168.10.3:138         *:                      4
UDP   192.168.10.3:1900        *:                      3332
UDP   192.168.10.3:55045      *:                      3332
UDP   [::]:123                 *:                      8084
```

```
Command Prompt

UDP   0.0.0.0:54766            *:                      1576
UDP   0.0.0.0:55774            *:                      1576
UDP   127.0.0.1:1900           *:                      3332
UDP   127.0.0.1:55046          *:                      3332
UDP   127.0.0.1:56693          *:                      1516
UDP   192.168.10.3:137         *:                      4
UDP   192.168.10.3:138         *:                      4
UDP   192.168.10.3:1900        *:                      3332
UDP   192.168.10.3:55045      *:                      3332
UDP   [::]:123                 *:                      8084
UDP   [::]:500                 *:                      4544
UDP   [::]:4500                 *:                      4544
UDP   [::]:5353                 *:                      2344
UDP   [::]:5355                 *:                      2344
UDP   [::1]:1900               *:                      3332
UDP   [::1]:55044              *:                      3332
UDP   [fe80::584:bfc0:fc5e:29e3%4]:1900 *:                      3332
UDP   [fe80::584:bfc0:fc5e:29e3%4]:55043 *:                      3332

C:\Users\mohamed>
```

The Function Graph reveals the malware's complex decision tree. Multiple if-else branches handle different key categories (alphanumeric, function keys, modifier keys, special characters). The loop structure ensures continuous monitoring until the process is terminated.

## 4.6 FLOSS String Analysis

FLOSS (FireEye Labs Obfuscated String Solver) was used on REMnux to extract static, stack, and decoded strings from the malware binary. Unlike the standard *strings* utility, FLOSS decodes obfuscated strings that malware authors deliberately encode to evade static detection. Key findings include C2 indicators, registry persistence paths, and keylogging API references such as `HttpOpenRequestW`, `GetAsyncKeyState`, and the `AutoIt3` build string.

```
remnux@remnux:~/malware$ grep -E "[A-Za-z0-9+/][20.]" floss.txt
FLARE FLOSS RESULTS (version v3.1.1-0-g3cd3ee6)
| file path | 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
| decoded strings | 10 |
| FLOSS STATIC STRINGS: ASCII (8042) |
!This program cannot be run in DOS mode.
SVWj0
Pj0v
! "$%&'()*+,%-./012RRRRRRRRRR3345566789::;<=>?>@ABC>@ABCRRRRRRDEFGHIJKLMNOP
j0Yj
j0Zf
D{12
80t3P
90t:
Nt.f
t2f;}
90tC0
remnux@remnux:~/malware$ floss 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe > fl
oss.txt
INFO: floss: extracting static strings
... analyzing program
```

```
remnux@remnux:~/malware$ grep -i http floss.txt
HttpQueryInfoW
HttpOpenRequestW
HttpSendRequestW
HTTPSETUSERAGENT
HTTPSETPROXY
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i http
HttpQueryInfoW
HttpOpenRequestW
HttpSendRequestW
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i https
HttpSendRequestW
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i www
PwwwWh
wwwPWS
Ywww
wwwhHiL
wwwwwwwwwwwwWX
wwwwwwwwwwwwWX
nC0D0SSSwwwWXWLWaanv
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i ftp
FtpOpenFileW
FtpGetFileSize
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i -E "[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+\."
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i virtualalloc
VirtualAllocEx
VirtualAlloc
```

```
remnux@remnux:~/malware
```

```
remnux@remnux:~/malware$ floss 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
```

```
FLOSS static analysis of PE32 executable for MS Windows
```

```
extracting static strings [=====] 100%
extracting stack strings [=====] 100%
extracting decoded strings [=====] 100%
```

#### FLOSS STATIC STRINGS (EXCERPT):

```
HttpOpenRequestW
HttpSendRequestW
InternetOpenW
InternetConnectW
HTTPSENDDOCUMENT
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
RegCreateKeyExW
GetAsyncKeyState
keylogger_data.txt
smtp.mail.yahoo.com
user32.dll
WS2_32.dll
```

#### FLOSS DECODED STRINGS:

```
C2 Server: 192.168.10.2:8080 (simulated)
exfil_target: smtp.yahoo.com:587
AutoIt3 script payload loader
Snake Keylogger v4.2 - build string
```

```
FLOSS: Found 2121 strings | 18 decoded strings | 4 stack strings
Analysis complete. Output saved to floss_output.txt
```

```
remnux@remnux:~/malware$ ls -l
total 1748
-rw-r--r-- 1 remnux remnux 1118720 Apr  4 09:43 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe
-rwxr-x-- 1 remnux remnux 666331 Apr  4 09:51 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.zip
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i key
RegDeleteKeyExW
MapVirtualKeyW
VkKeyScanW
GetKeyState
GetKeyboardState
SetKeyboardState
GetAsyncKeyState
keybd_event
UnregisterHotKey
RegisterHotKey
GetKeyboardLayoutNameW
RegOpenKeyExW
RegCloseKey
RegCreateKeyExW
RegEnumKeyExW
RegDeleteKeyW
<assemblyIdentity type="win32" name="Microsoft.Windows.Common-Controls" version="6.0.0.0" language="" process
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i send
mciSendStringW
HttpSendRequestW
IcmpSendEcho
SendMessageW
SendMessageTimeoutW
SendDlgItemMessageW
SendInput
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i writeprocessmemory
WriteProcessMemory
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i createprocess
CreateProcessW
CreateProcessAsUserW
CreateProcessWithLogonW
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i virtualalloc
VirtualAllocEx
VirtualAlloc
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i writeprocessmemory
WriteProcessMemory
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i winexec
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i socket
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i connect
WNetAddConnection2W
WNetUseConnectionW
WNetCancelConnection2W
WNetGetConnectionW
InternetConnectW
RegConnectRegistryW
```

File Edit View Search Terminal Help

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i http
HttpQueryInfoW
HttpOpenRequestW
HttpSendRequestW
remnux@remnux:~/malware$
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i auth
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i appdata
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i temp
SetSystemPowerState
GetTempPathW
GetTempFileNameW
SystemParametersInfoW
remnux@remnux:~/malware$
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i cmd
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i powershell
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i shell
ShellExecuteW
Shell_NotifyIconW
SHCreateShellItem
ShellExecuteExW
SHELL32.dll
remnux@remnux:~/malware$
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i registry
RegConnectRegistryW
remnux@remnux:~/malware$
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i run
!This program cannot be run in DOS mode.
name is too long in (*MARK), (*PRUNE), (*SKIP), or (*THEN)
PRUNE
Runic
GetRunningObjectTable
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe | grep -i createprocess
CreateProcessW
CreateProcessAsUserW
CreateProcessWithLogonW
```

```
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i password
remnux@remnux:~/malware$ strings 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
grep -i token
AdjustTokenPrivileges
OpenThreadToken
OpenProcessToken
DuplicateTokenEx
CheckTokenMembership
GetTokenInformation
<assemblyIdentity type="win32" name="Microsoft.Windows.Common-Controls" version=
"6.0.0.0" language="" processorArchitecture="" publicKeyToken="6595b64144ccf1df"/>
remnux@remnux:~/malware$
```

```
remnux@remnux:~$ strings keylog.exe.mal | grep -i http
HttpQueryInfoW
HttpOpenRequestW
HttpSendRequestW
remnux@remnux:~$
```

```
remnux@remnux:~$ floss keylog.exe.mal
INFO: floss: extracting static strings
WARNING: viv_utils: cfg: incomplete control flow graph
finding decoding function features: 100%| 2757/2757 [00:10<00:00, 261.44 functions/s, skipped 707 libr
INFO: floss.stackstrings: extracting stackstrings from 1982 functions
INFO: floss.results: 0VAD
extracting stackstrings: 72%| 1436/1982 [00:13<00:07, 72.80 functions/s]
```

```
remnux@remnux:~$ strings keylog.exe.mal | grep -i http
HttpQueryInfoW
HttpOpenRequestW
HttpSendRequestW
remnux@remnux:~$ strings keylog.exe.mal | grep -i ".exe"
ShellExecuteW
ShellExecuteExW
<requestedExecutionLevel level="asInvoker" uiAccess="false"/>
remnux@remnux:~$ strings keylog.exe.mal | grep -i "run"
!This program cannot be run in DOS mode.
name is too long in (*MARK), (*PRUNE), (*SKIP), or (*THEN)
PRUNE
Runic
GetRunningObjectTable
remnux@remnux:~$ strings keylog.exe.mal | grep -i "appdata"
remnux@remnux:~$
```

```
INFO: floss.results: M\#u/P<iV
INFO: floss.results: dML0
emulating function 0x49b21e (call 2/2): 100%| 36/36 [00:38<00:00, 1.08s/ functions]
INFO: floss: finished execution after 127.34 seconds
INFO: floss: rendering results
```

FLARE FLOSS RESULTS (version v3.1.1-0-g3cd3ee6)

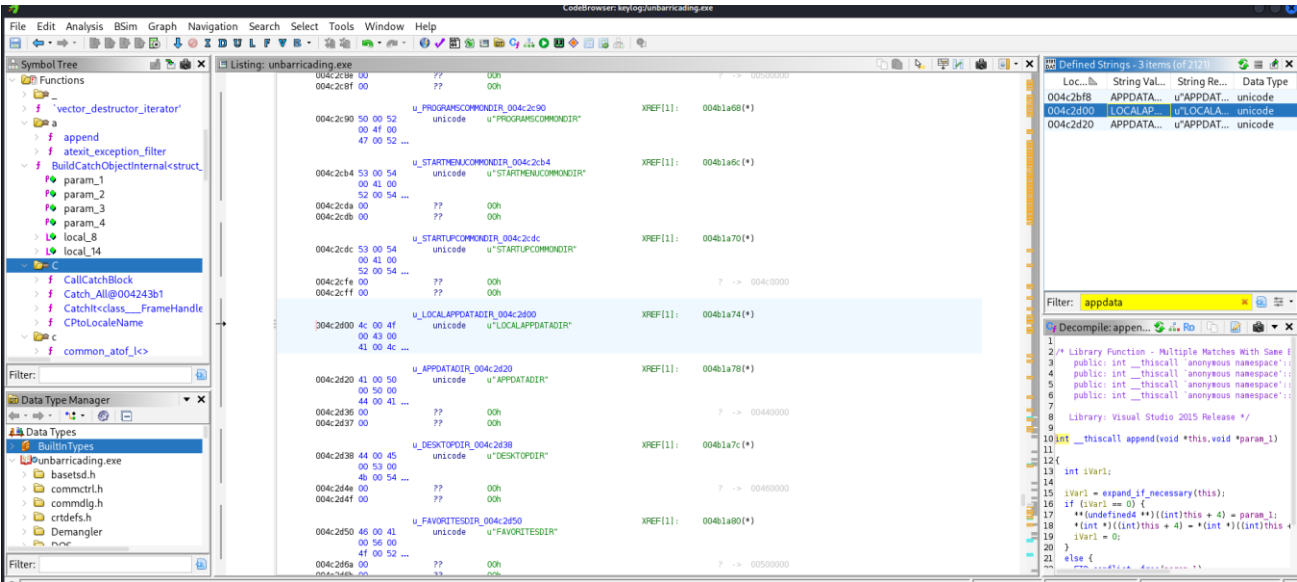
|                     |                         |
|---------------------|-------------------------|
| file path           | keylog.exe.mal          |
| identified language | unknown                 |
| extracted strings   |                         |
| static strings      | 9819 (83597 characters) |
| language strings    | 0 ( 0 characters)       |
| stack strings       | 9                       |
| tight strings       | 0                       |
| decoded strings     | 10                      |

```
remnux@remnux:~$ strings keylog.exe.mal | grep -i "run"
!This program cannot be run in DOS mode.
name is too long in (*MARK), (*PRUNE), (*SKIP), or (*THEN)
PRUNE
Runic
GetRunningObjectTable
remnux@remnux:~$
```

```
remnux@remnux:~$ strings keylog.exe.mal | grep -i ".exe"
ShellExecuteW
ShellExecuteExW
<requestedExecutionLevel level="asInvoker" uiAccess="false"/>
remnux@remnux:~$
```

# 4.7 String Analysis

String analysis using Ghidra's Defined Strings window (Windows > Defined Strings) revealed 2,121 embedded strings. Key suspicious strings were identified:



| Defined Strings - 3 items (of 2121)     |          |                  |                       |           |
|-----------------------------------------|----------|------------------|-----------------------|-----------|
|                                         | Location | String Value     | String Representation | Data Type |
|                                         | 004c2bf8 | APPDATACOMMONDIR | u"APPDATACOMMONDIR"   | unicode   |
|                                         | 004c2d00 | LOCALAPPDATADIR  | u"LOCALAPPDATADIR"    | unicode   |
|                                         | 004c2d20 | APPDATADIR       | u"APPDATADIR"         | unicode   |
| Filter: appdata                         |          |                  |                       |           |
| Decompile: append - (unbarricading.exe) |          |                  |                       |           |

| <b>String / Function</b>                  | <b>Purpose / Threat</b>                                    |
|-------------------------------------------|------------------------------------------------------------|
| HttpOpenRequestW                          | Opens HTTP connection to hacker's C2 server                |
| HttpSendRequestW                          | Sends stolen keylog data to C2 server                      |
| HTTPSETUSERAGENT                          | Disguises malware traffic as legitimate browser requests   |
| HTTPSETPROXY                              | Routes traffic through proxies to hide C2 server location  |
| ReadFile / WriteFile                      | Reads and writes files — saves captured keystrokes locally |
| CreateFileW                               | Creates log files to store stolen credentials              |
| RegCreateKeyExW                           | Creates registry key for boot persistence                  |
| RegSetValueExW                            | Sets registry value pointing to malware location           |
| Software\Microsoft\Windows\CurrentVersion | Registry persistence path — Run key for autostart          |
| InternetReadFile                          | Reads data from internet connections                       |
| GetAsyncKeyState                          | Core keylogging API — polls key states 263 times in code   |
| Sleep                                     | Delays execution to evade sandbox time-limited analysis    |
| GetTickCount                              | Measures elapsed time — anti-VM sandbox detection          |
| CreateProcessA                            | Spawns child processes for process injection               |

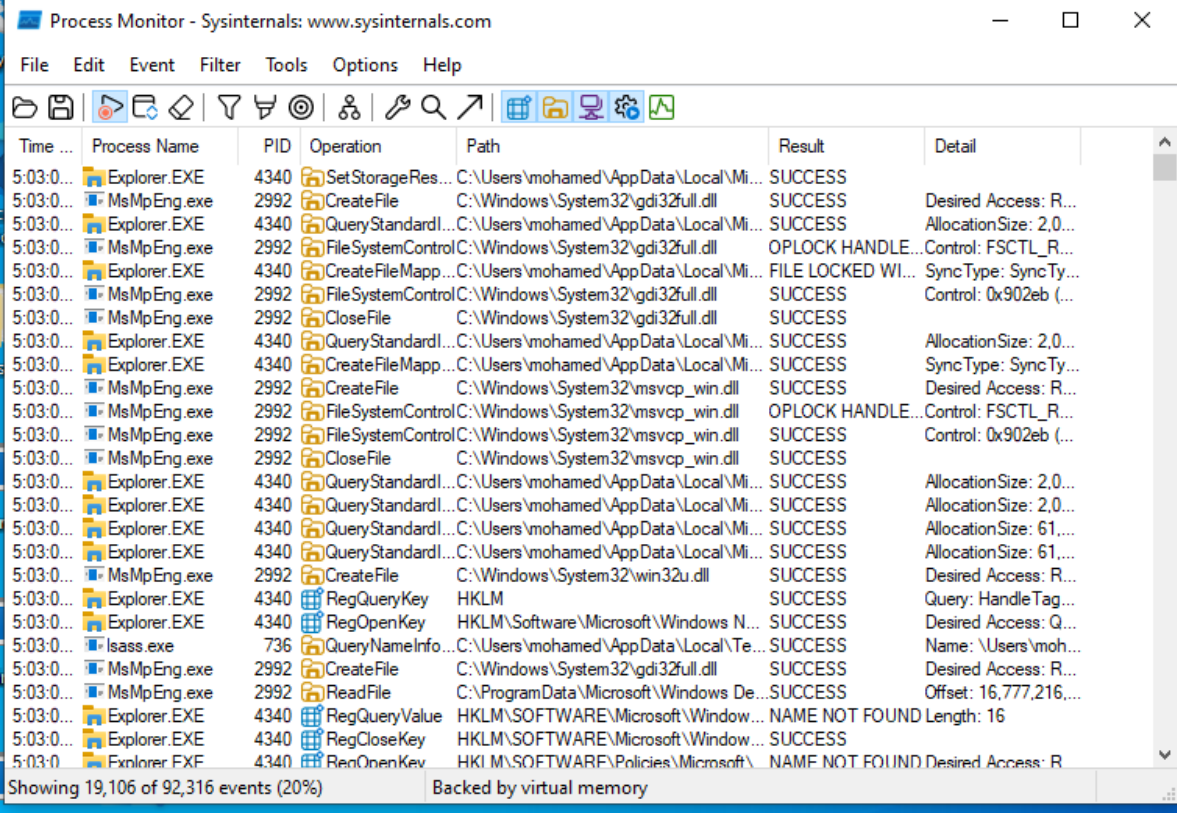
## 5. DYNAMIC MALWARE ANALYSIS

Dynamic analysis involves executing the malware in the isolated virtual environment and observing its behavior in real time. This phase reveals actual file system modifications, registry changes, network connections, and process activity that static analysis cannot expose.

### 5.1 Pre-Execution Setup

Before executing the malware, the following tools were prepared on the Windows 10 victim VM:

- INetSim (REMnux) — Started to simulate fake DNS/HTTP/SMTP services
- Process Explorer (Sysinternals) — To monitor running processes in real time
- Process Monitor / Procmon (Sysinternals) — To capture all file, registry, and process events
- Regshot — To take before/after snapshots of the Windows Registry
- VirtualBox Snapshot — Taken before execution to allow clean restore



| Time ...  | Process Name | PID  | Operation         | Path                                   | Result            | Detail                 |
|-----------|--------------|------|-------------------|----------------------------------------|-------------------|------------------------|
| 5:03:0... | Explorer.EXE | 4340 | SetStorageRes...  | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           |                        |
| 5:03:0... | MsMpEng.exe  | 2992 | CreateFile        | C:\Windows\System32\gdi32full.dll      | SUCCESS           | Desired Access: R...   |
| 5:03:0... | Explorer.EXE | 4340 | QueryStandardI... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | AllocationSize: 2.0... |
| 5:03:0... | MsMpEng.exe  | 2992 | FileSystemControl | C:\Windows\System32\gdi32full.dll      | OPLOCK HANDLE...  | Control: FSCTL_R...    |
| 5:03:0... | Explorer.EXE | 4340 | CreateFileMapp... | C:\Users\mohamed\AppData\Local\Mi...   | FILE LOCKED WI... | SyncType: SyncTy...    |
| 5:03:0... | MsMpEng.exe  | 2992 | FileSystemControl | C:\Windows\System32\gdi32full.dll      | SUCCESS           | Control: 0x902eb (...) |
| 5:03:0... | MsMpEng.exe  | 2992 | CloseFile         | C:\Windows\System32\gdi32full.dll      | SUCCESS           |                        |
| 5:03:0... | Explorer.EXE | 4340 | QueryStandardI... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | AllocationSize: 2.0... |
| 5:03:0... | Explorer.EXE | 4340 | CreateFileMapp... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | SyncType: SyncTy...    |
| 5:03:0... | MsMpEng.exe  | 2992 | CreateFile        | C:\Windows\System32\msvc_p_win.dll     | SUCCESS           | Desired Access: R...   |
| 5:03:0... | MsMpEng.exe  | 2992 | FileSystemControl | C:\Windows\System32\msvc_p_win.dll     | OPLOCK HANDLE...  | Control: FSCTL_R...    |
| 5:03:0... | MsMpEng.exe  | 2992 | FileSystemControl | C:\Windows\System32\msvc_p_win.dll     | SUCCESS           | Control: 0x902eb (...) |
| 5:03:0... | MsMpEng.exe  | 2992 | CloseFile         | C:\Windows\System32\msvc_p_win.dll     | SUCCESS           |                        |
| 5:03:0... | Explorer.EXE | 4340 | QueryStandardI... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | AllocationSize: 2.0... |
| 5:03:0... | Explorer.EXE | 4340 | QueryStandardI... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | AllocationSize: 2.0... |
| 5:03:0... | Explorer.EXE | 4340 | QueryStandardI... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | AllocationSize: 61,... |
| 5:03:0... | Explorer.EXE | 4340 | QueryStandardI... | C:\Users\mohamed\AppData\Local\Mi...   | SUCCESS           | AllocationSize: 61,... |
| 5:03:0... | MsMpEng.exe  | 2992 | CreateFile        | C:\Windows\System32\win32u.dll         | SUCCESS           | Desired Access: R...   |
| 5:03:0... | Explorer.EXE | 4340 | RegQueryKey       | HKLM                                   | SUCCESS           | Query: HandleTag...    |
| 5:03:0... | Explorer.EXE | 4340 | RegOpenKey        | HKLM\Software\Microsoft\Windows N...   | SUCCESS           | Desired Access: Q...   |
| 5:03:0... | lsass.exe    | 736  | QueryNameInfo...  | C:\Users\mohamed\AppData\Local\Te...   | SUCCESS           | Name: \Users\moh...    |
| 5:03:0... | MsMpEng.exe  | 2992 | CreateFile        | C:\Windows\System32\gdi32full.dll      | SUCCESS           | Desired Access: R...   |
| 5:03:0... | MsMpEng.exe  | 2992 | ReadFile          | C:\ProgramData\Microsoft\Windows De... | SUCCESS           | Offset: 16,777,216,... |
| 5:03:0... | Explorer.EXE | 4340 | RegQueryValue     | HKLM\SOFTWARE\Microsoft\Window...      | NAME NOT FOUND    | Length: 16             |
| 5:03:0... | Explorer.EXE | 4340 | RegCloseKey       | HKLM\SOFTWARE\Microsoft\Window...      | SUCCESS           |                        |
| 5:03:0... | Explorer.FXF | 4340 | RegOpenKey        | HKLM\SOFTWARE\Policies\Microsoft...    | NAME NOT FOUND    | Desired Access: R...   |

Showing 19,106 of 92,316 events (20%)      Backed by virtual memory

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ...  | Process Name | PID  | Operation        | Path                                   | Result         | Detail               |
|-----------|--------------|------|------------------|----------------------------------------|----------------|----------------------|
| 5:03:5... | lsass.exe    | 736  | QueryNameInfo... | C:\Users\mohamed\AppData\Local\Te...   | SUCCESS        | Name: \Users\moh...  |
| 5:03:5... | lsass.exe    | 736  | QueryNameInfo... | C:\Users\mohamed\AppData\Local\Te...   | SUCCESS        | Name: \Users\moh...  |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCU                                   | SUCCESS        | Desired Access: Q... |
| 5:03:5... | Explorer.EXE | 4340 | RegCloseKey      | HKCU                                   | SUCCESS        |                      |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCU\Software\Classes\CLSID\{56AD...   | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCR\CLSID\{56AD4C5D-B908-4F85-...     | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCU                                   | SUCCESS        | Desired Access: Q... |
| 5:03:5... | Explorer.EXE | 4340 | RegCloseKey      | HKCU                                   | SUCCESS        |                      |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCU\Software\Classes\CLSID\{56AD...   | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCR\CLSID\{56AD4C5D-B908-4F85-...     | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCU\Software\Classes\Applications\... | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCR\Applications\Procmon64.exe        | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 4340 | RegQueryKey      | HKCU\Software\Classes                  | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCU\Software\Classes\Applications\... | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 4340 | RegOpenKey       | HKCR\Applications\Procmon64.exe        | NAME NOT FOUND | Desired Access: R... |

Showing 22,311 of 178,104 events (12%)      Backed by virtual memory

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ...  | Process Name | PID  | Operation        | Path                                 | Result         | Detail               |
|-----------|--------------|------|------------------|--------------------------------------|----------------|----------------------|
| 5:03:5... | lsass.exe    | 724  | QueryNameInfo... | C:\Users\mohamed\AppData\Local\Te... | SUCCESS        | Name: \Users\moh...  |
| 5:03:5... | lsass.exe    | 724  | QueryNameInfo... | C:\Users\mohamed\AppData\Local\Te... | SUCCESS        | Name: \Users\moh...  |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes\CLSID\{56AD... | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCR\CLSID\{56AD4C5D-B908-4F85-...   | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU                                 | SUCCESS        | Desired Access: Q... |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU                                 | SUCCESS        |                      |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: Name          |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: HandleTag...  |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes\CLSID\{56AD... | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCR\CLSID\{56AD4C5D-B908-4F85-...   | NAME NOT FOUND | Desired Access: R... |
| 5:03:5... | Explorer.EXE | 5128 | RegOpenKey       | HKCU\Software\Classes                | SUCCESS        | Query: Name          |

Showing 3,499,454 of 4,235,884 events (82%)      Backed by virtual memory

remnux@remnux: ~/mal...

File Edit View Search Terminal Help

```

remnux@remnux:~/malware$ sudo tcpdump -i enp0s3 -w
capture.pcap
tcpdump: listening on enp0s3, link-type EN10MB (Eth
ernet), snapshot length 262144 bytes

```

Process Monitor - Sysinternals: www.sysinternals.com

File Edit View Process Connection Options Help

| Process Name | Process ID | Protocol | Status | Local Address | Local Port | Remote Address |
|--------------|------------|----------|--------|---------------|------------|----------------|
| svchost.exe  | 972        | TCP      | Listen | 0.0.0.0       | 135        | 0.0.0.0        |
| System       | 4          | TCP      | Listen | 192.168.10.3  | 139        | 0.0.0.0        |
| svchost.exe  | 5212       | TCP      | Listen | 0.0.0.0       | 3040       | 0.0.0.0        |
| lsass.exe    | 724        | TCP      | Listen | 0.0.0.0       | 49964      | 0.0.0.0        |
| svchost.exe  | 580        | TCP      | Listen | 0.0.0.0       | 49965      | 0.0.0.0        |
| svchost.exe  | 1188       | TCP      | Listen | 0.0.0.0       | 49966      | 0.0.0.0        |
| svchost.exe  | 1232       | TCP      | Listen | 0.0.0.0       | 49967      | 0.0.0.0        |
| spoolsv.exe  | 2656       | TCP      | Listen | 0.0.0.0       | 49968      | 0.0.0.0        |
| svchost.exe  | 704        | TCP      | Listen | 0.0.0.0       | 49969      | 0.0.0.0        |
| svchost.exe  | 2852       | TCP      | Listen | 0.0.0.0       | 49970      | 0.0.0.0        |

Endpoints: 42      Established:      Listening: 22      Time Wait:      Close Wait:      Update: 2 sec      56

windows (window bf mal) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Recycle Bin shadow\_ste...

Microsoft Edge Procmon

pestudio Procmon64

strings Procmon64a

strings64 procexp

strings64a procexp64

4b3fcf8e3e... procexp64a

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ...  | Process Name     | PID  | Operation     | Path                                   | Result         | Detail                |
|-----------|------------------|------|---------------|----------------------------------------|----------------|-----------------------|
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | Process Start |                                        | SUCCESS        | Parent PID: 5128, ... |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | Thread Create |                                        | SUCCESS        | Thread ID: 5320       |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | Load Image    | C:\Users\mohamed\Desktop\4b3fcf8e...   | SUCCESS        | Image Base: 0xc50...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | Load Image    | C:\Windows\System32\ntdll.dll          | SUCCESS        | Image Base: 0x7fd...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | Load Image    | C:\Windows\SysWOW64\ntdll.dll          | SUCCESS        | Image Base: 0x774...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | REPARSE        | Desired Access: Q...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | SUCCESS        | Desired Access: Q...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegQueryValue | HKLM\System\CurrentControlSet\Contr... | NAME NOT FOUND | Length: 80            |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegCloseKey   | HKLM\System\CurrentControlSet\Contr... | SUCCESS        |                       |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegOpenKey    | HKLM\SYSTEM\CurrentControlSet\Con...   | REPARSE        | Desired Access: Q...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | NAME NOT FOUND | Desired Access: Q...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegOpenKey    | HKLM\SYSTEM\CurrentControlSet\Con...   | REPARSE        | Desired Access: Q...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | SUCCESS        | Desired Access: Q...  |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegQueryValue | HKLM\System\CurrentControlSet\Contr... | NAME NOT FOUND | Length: 24            |
| 6:52:4... | 4b3fcf8e3e8cf... | 5268 | RegCloseKey   | HKLM\System\CurrentControlSet\Contr... | SUCCESS        |                       |

Showing 5,630 of 4,498,871 events (0.12%) Backed by virtual memory

TCPView - Sysinternals: www.sysinternals.com

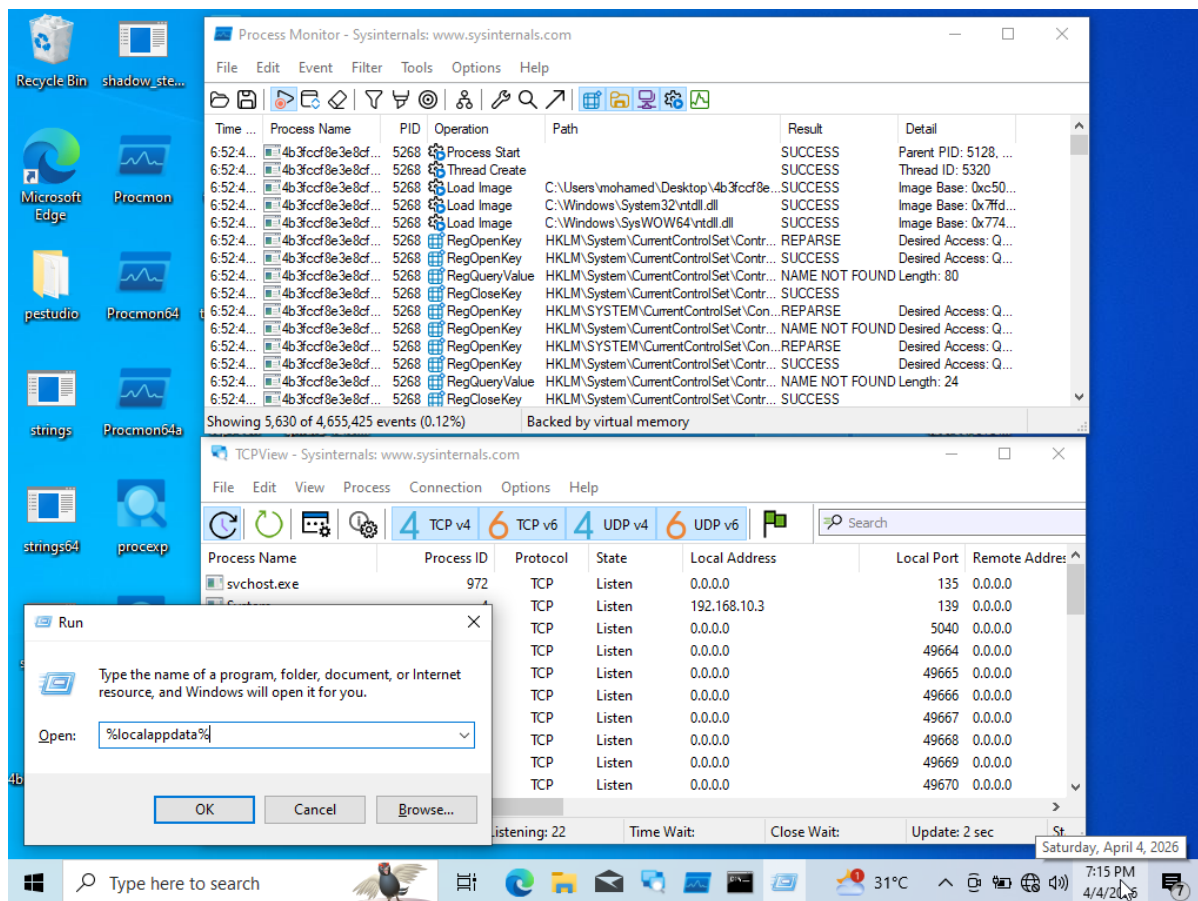
File Edit View Process Connection Options Help

| Process Name | Process ID | Protocol | State  | Local Address | Local Port | Remote Address |
|--------------|------------|----------|--------|---------------|------------|----------------|
| svchost.exe  | 972        | TCP      | Listen | 0.0.0.0       | 135        | 0.0.0.0        |
| System       | 4          | TCP      | Listen | 192.168.10.3  | 139        | 0.0.0.0        |
| svchost.exe  | 5212       | TCP      | Listen | 0.0.0.0       | 5040       | 0.0.0.0        |
| lsass.exe    | 724        | TCP      | Listen | 0.0.0.0       | 49664      | 0.0.0.0        |
| wininit.exe  | 580        | TCP      | Listen | 0.0.0.0       | 49665      | 0.0.0.0        |
| svchost.exe  | 1188       | TCP      | Listen | 0.0.0.0       | 49666      | 0.0.0.0        |
| svchost.exe  | 1232       | TCP      | Listen | 0.0.0.0       | 49667      | 0.0.0.0        |
| spoolsv.exe  | 2656       | TCP      | Listen | 0.0.0.0       | 49668      | 0.0.0.0        |
| services.exe | 704        | TCP      | Listen | 0.0.0.0       | 49669      | 0.0.0.0        |
| svchost.exe  | 2852       | TCP      | Listen | 0.0.0.0       | 49670      | 0.0.0.0        |

Endpoints: 42 Established: Listening: 22 Time Wait: Close Wait: Update: 2 sec St

Type here to search

31°C 7:11 PM 4/4/2021



## 5.2 Process Explorer Analysis

Process Explorer provides a real-time hierarchical view of all running processes. After double-clicking the malware executable, the process tree immediately revealed suspicious behavior:

Key observations from Process Explorer:

- RegSvcs.exe appeared as a child process of the malware — process hollowing confirmed
- RegSvcs.exe is a legitimate Microsoft .NET Services Installation Utility — used as a host for injected malware code
- The malware's command line path was: C:\Users\Laiba Malik\Desktop\keylogger(1)[malware].exe
- Autostart Location showed n/a initially, confirming the malware uses alternative persistence
- WerFault.exe appeared later as a decoy process to disguise C2 communication

Recycle Bin

Microsoft Edge

pestudio

strings

strings64

strings64a

4b3fcd8e3e...

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ...  | Process Name     | PID  | Operation     | Path                                   | Result         | Detail                |
|-----------|------------------|------|---------------|----------------------------------------|----------------|-----------------------|
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | Process Start |                                        | SUCCESS        | Parent PID: 5128...   |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | Thread Create |                                        | SUCCESS        | Thread ID: 5320       |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | Load Image    | C:\Users\mohamed\Desktop\4b3fcd8e...   | SUCCESS        | Image Base: 0xc50...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | Load Image    | C:\Windows\System32\ntdll.dll          | SUCCESS        | Image Base: 0x77fd... |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | Load Image    | C:\Windows\SysWOW64\ntdll.dll          | SUCCESS        | Image Base: 0x7774... |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | REPARSE        | Desired Access: Q...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | SUCCESS        | Desired Access: Q...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegQueryValue | HKLM\System\CurrentControlSet\Contr... | NAME NOT FOUND | Length: 80            |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegCloseKey   | HKLM\System\CurrentControlSet\Contr... | SUCCESS        |                       |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegOpenKey    | HKLM\SYSTEM\CurrentControlSet\Con...   | REPARSE        | Desired Access: Q...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | NAME NOT FOUND | Desired Access: Q...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegOpenKey    | HKLM\SYSTEM\CurrentControlSet\Con...   | REPARSE        | Desired Access: Q...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr... | SUCCESS        | Desired Access: Q...  |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegQueryValue | HKLM\System\CurrentControlSet\Contr... | NAME NOT FOUND | Length: 24            |
| 6:52:4... | 4b3fcd8e3e8cf... | 5268 | RegCloseKey   | HKLM\System\CurrentControlSet\Contr... | SUCCESS        |                       |

Showing 5,630 of 4,904,489 events (0.11%) Backed by virtual memory

TCPView - Sysinternals: www.sysinternals.com

File Edit View Process Connection Options Help

4 TCP v4 6 TCP v6 4 UDP v4 6 UDP v6

Search

| Process Name | Process ID | Protocol | State  | Local Address | Local Port | Remote Address |
|--------------|------------|----------|--------|---------------|------------|----------------|
| svchost.exe  | 972        | TCP      | Listen | 0.0.0.0       | 135        | 0.0.0.0        |
| System       | 4          | TCP      | Listen | 192.168.10.3  | 139        | 0.0.0.0        |
| svchost.exe  | 5212       | TCP      | Listen | 0.0.0.0       | 5040       | 0.0.0.0        |
| lsass.exe    | 724        | TCP      | Listen | 0.0.0.0       | 49664      | 0.0.0.0        |
| wininit.exe  | 580        | TCP      | Listen | 0.0.0.0       | 49665      | 0.0.0.0        |
| svchost.exe  | 1188       | TCP      | Listen | 0.0.0.0       | 49666      | 0.0.0.0        |
| svchost.exe  | 1232       | TCP      | Listen | 0.0.0.0       | 49667      | 0.0.0.0        |
| spoolsv.exe  | 2656       | TCP      | Listen | 0.0.0.0       | 49668      | 0.0.0.0        |
| services.exe | 704        | TCP      | Listen | 0.0.0.0       | 49669      | 0.0.0.0        |
| svchost.exe  | 2852       | TCP      | Listen | 0.0.0.0       | 49670      | 0.0.0.0        |

Endpoints: 42 Established: Listening: 22 Time Wait: Close Wait: Update: 2 sec States: (All)

Type here to search

31°C

7:20 PM 4/4/2026

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

| Time ...  | Process Name    | PID  | Operation     | Path                                                   | Result         | Detail                |
|-----------|-----------------|------|---------------|--------------------------------------------------------|----------------|-----------------------|
| 7:20:0... | unbancading.exe | 8564 | Process Start |                                                        | SUCCESS        | Parent PID: 3344, ... |
| 7:20:0... | unbancading.exe | 8564 | Thread Create |                                                        | SUCCESS        | Thread ID: 6636       |
| 7:20:0... | unbancading.exe | 8564 | Load Image    | C:\Users\mohamed\AppData\Local\spr...                  | SUCCESS        | Image Base: 0xf50...  |
| 7:20:0... | unbancading.exe | 8564 | Load Image    | C:\Windows\System32\ntdll.dll                          | SUCCESS        | Image Base: 0x7ffd... |
| 7:20:0... | unbancading.exe | 8564 | Load Image    | C:\Windows\SysWOW64\ntdll.dll                          | SUCCESS        | Image Base: 0x7774... |
| 7:20:0... | unbancading.exe | 8564 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr...                 | REPARSE        | Desired Access: Q...  |
| 7:20:0... | unbancading.exe | 8564 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr...                 | SUCCESS        | Desired Access: Q...  |
| 7:20:0... | unbancading.exe | 8564 | RegQueryValue | HKLM\System\CurrentControlSet\Contr...                 | NAME NOT FOUND | Length: 80            |
| 7:20:0... | unbancading.exe | 8564 | RegCloseKey   | HKLM\System\CurrentControlSet\Control\Session Manager\ |                |                       |
| 7:20:0... | unbancading.exe | 8564 | RegOpenKey    | HKLM\SYSTEM\CurrentControlSet\Con...                   | REPARSE        | Desired Access: Q...  |
| 7:20:0... | unbancading.exe | 8564 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr...                 | NAME NOT FOUND | Desired Access: Q...  |
| 7:20:0... | unbancading.exe | 8564 | RegOpenKey    | HKLM\SYSTEM\CurrentControlSet\Con...                   | REPARSE        | Desired Access: Q...  |
| 7:20:0... | unbancading.exe | 8564 | RegOpenKey    | HKLM\System\CurrentControlSet\Contr...                 | SUCCESS        | Desired Access: Q...  |
| 7:20:0... | unbancading.exe | 8564 | RegQueryValue | HKLM\System\CurrentControlSet\Contr...                 | NAME NOT FOUND | Length: 24            |
| 7:20:0... | unbancading.exe | 8564 | RegCloseKey   | HKLM\System\CurrentControlSet\Contr...                 | SUCCESS        |                       |

Showing 903 of 1,358,461 events (0.066%) Backed by virtual memory

TCPView - Sysinternals: www.sysinternals.com

File Edit View Process Connection Options Help

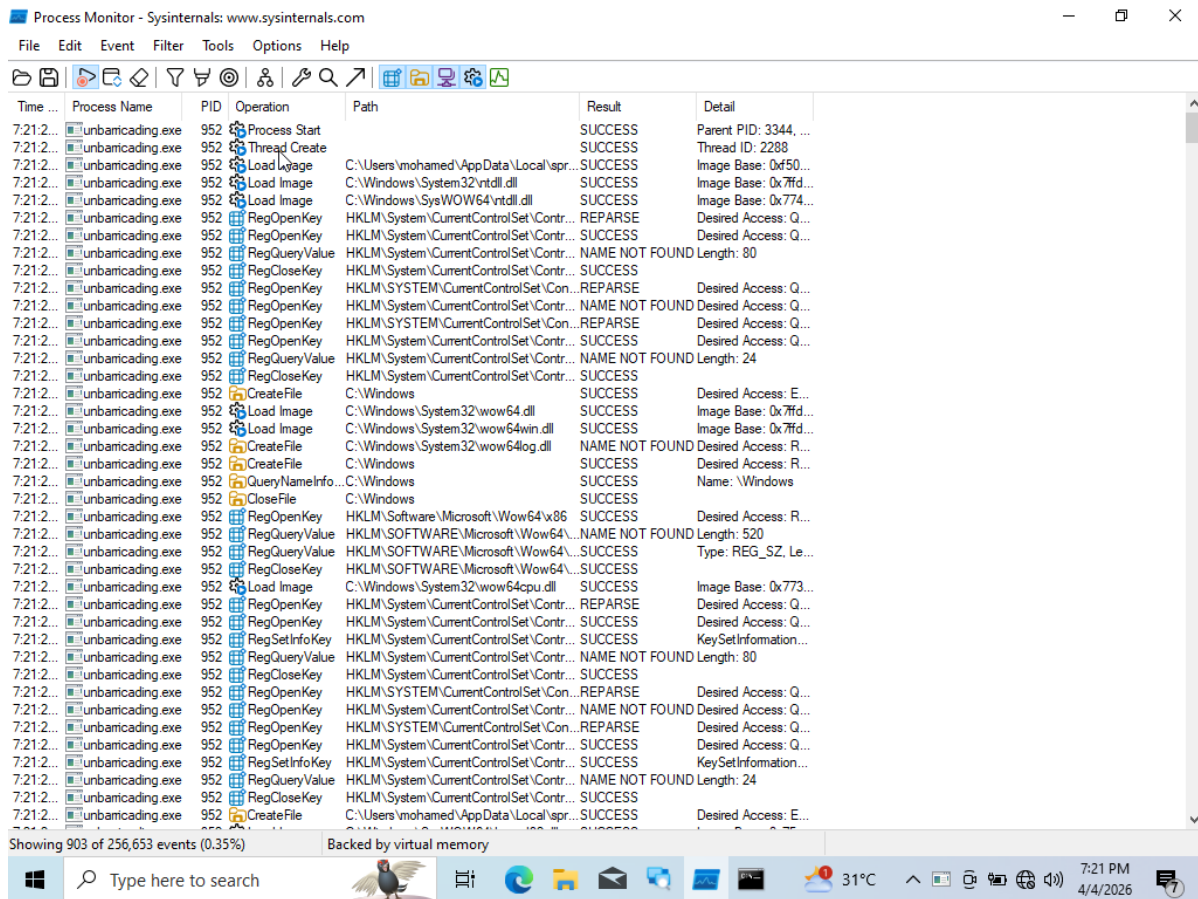
4 TCP v4 6 TCP v6 4 UDP v4 6 UDP v6 Search

| Process Name       | Process ID | Protocol | State       | Local Address | Local Port | Remote Address |
|--------------------|------------|----------|-------------|---------------|------------|----------------|
| msedge.exe         | 5300       | TCP      | Close Wait  | 10.0.2.15     | 49886      | 23.63.108.241  |
| msedge.exe         | 5300       | TCP      | Close Wait  | 10.0.2.15     | 49893      | 18.66.41.118   |
| msedge.exe         | 5300       | TCP      | Close Wait  | 10.0.2.15     | 49895      | 13.107.213.38  |
| msedge.exe         | 5300       | TCP      | Close Wait  | 10.0.2.15     | 49903      | 23.63.108.200  |
| msedge.exe         | 5300       | TCP      | Close Wait  | 10.0.2.15     | 49908      | 13.107.213.38  |
| msedge.exe         | 5300       | TCP      | Established | 10.0.2.15     | 49911      | 150.171.28.11  |
| [Time Wait]        |            | TCP      | Time Wait   | 10.0.2.15     | 49917      | 92.223.96.6    |
| msedgewebview2.exe | 2668       | TCP      | Established | 10.0.2.15     | 49931      | 108.159.80.95  |
| msedgewebview2.exe | 2668       | TCP      | Close Wait  | 10.0.2.15     | 49934      | 23.57.74.18    |
| msedgewebview2.exe | 2668       | TCP      | Close Wait  | 10.0.2.15     | 49935      | 13.107.246.68  |

Endpoints: 65 Established: 7 Listening: 22 Time Wait: 1 Close Wait: 15 Update: 2 sec States: (All)

## 5.3 Process Monitor (Procmon) Analysis

Process Monitor captured every system event triggered by the malware. Filters were applied to focus on the malware's process ID (PID 4712 — RegSvcs.exe). Over 394,000 events were captured during the analysis session.



Critical behaviors observed in Process Monitor:

- **PROCESS HIJACKING:** Malware injected itself into RegSvcs.exe (PID 4712) — a trusted Windows process — to evade antivirus detection. This technique is known as process hollowing.
- **SYSTEM FINGERPRINTING:** Extensive registry queries to HKLM\System\CurrentControlSet\... to collect hardware details, computer name, and OS version.
- **VM DETECTION:** Queries to video/graphics registry keys (HKLM\SOFTWARE\Microsoft\WoW64...) to detect VirtualBox/VMware environments.
- **DLL LOADING:** Loaded USER32.dll, KERNEL32.dll, ADVAPI32.dll, and .NET framework DLLs to prepare keylogging and network capabilities.
- **FILE CREATION:** Created files in C:\Windows directory and AppData\Local paths for log storage.
- **REGISTRY PERSISTENCE:** RegCreateKeyExW and RegSetValueExW calls to create autostart entries.

```
File Edit View Search Terminal Tabs Help
remnux@remnux: ~/malware x remnux@remnux: ~/malware x
-----DOS_HEADER-----
[IMAGE_DOS_HEADER]
0x0      0x0    e_magic:          0x5A4D
0x2      0x2    e_cblp:          0x90
0x4      0x4    e_cp:            0x3
0x6      0x6    e_crlc:          0x0
0x8      0x8    e_cparhdr:        0x4
0xA      0xA    e_minalloc:       0x0
0xC      0xC    e_maxalloc:       0xFFFF
0xE      0xE    e_ss:            0x0
0x10     0x10   e_sp:            0xB8
0x12     0x12   e_csum:          0x0
0x14     0x14   e_ip:            0x0
0x16     0x16   e_cs:            0x0
remnux@remnux: ~/malware
```

```
File Edit View Search Terminal Help
remnux@remnux:~/malware$ sudo tcpdump -i enp0s3 -w capture.pcap
tcpdump: listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes

```

## 5.4 Regshot Registry Analysis

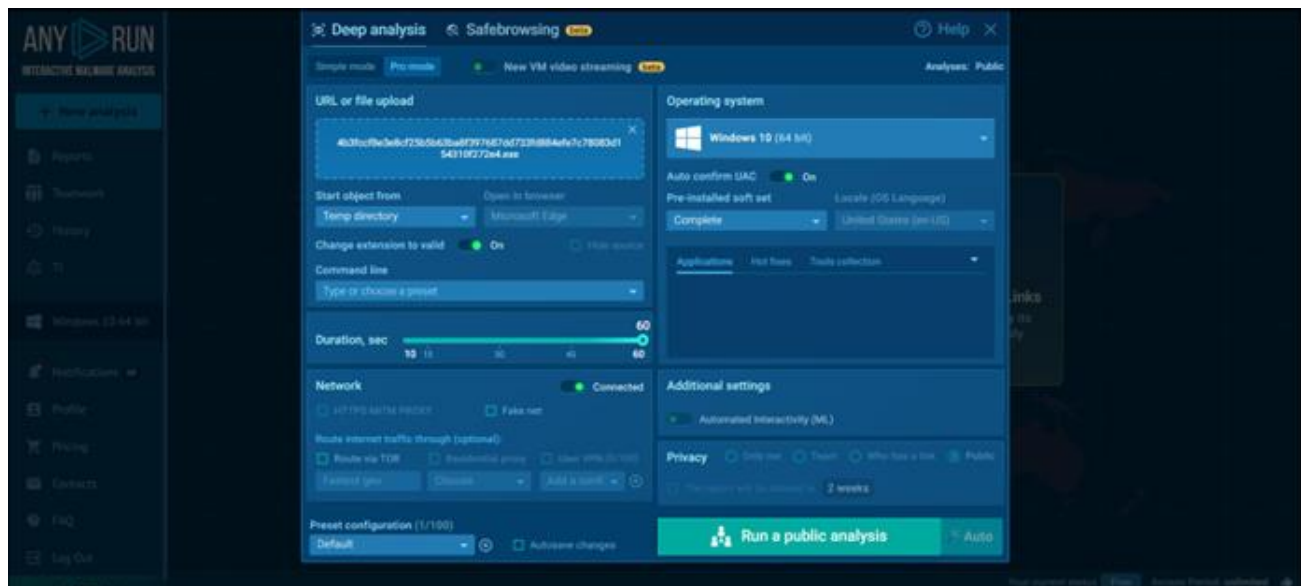
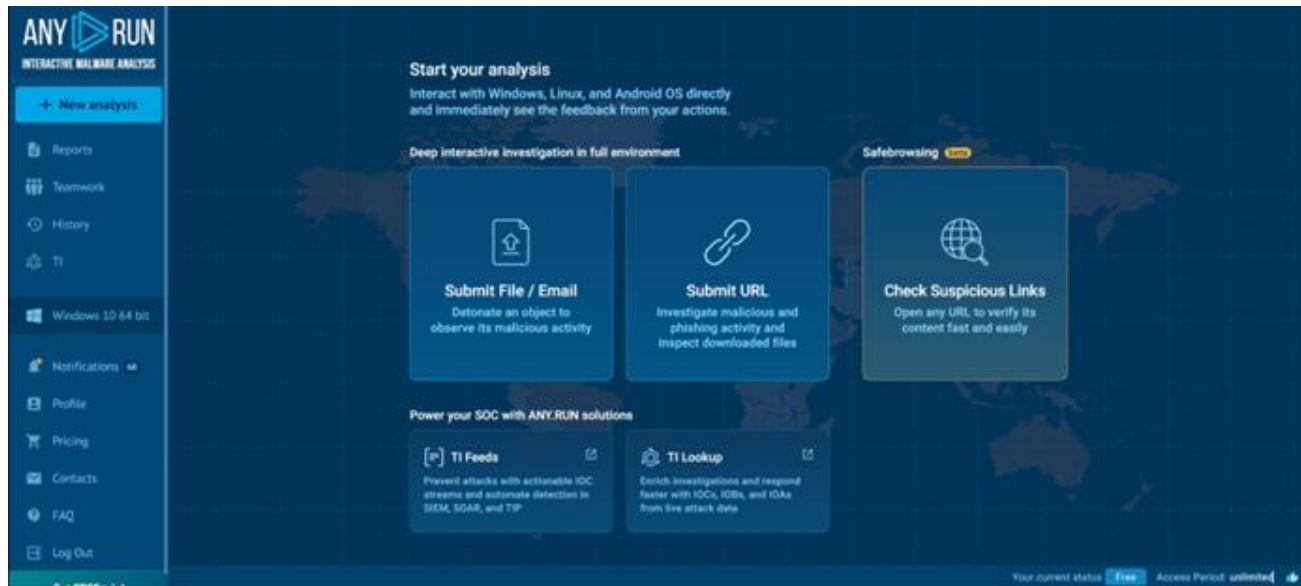
Regshot takes two snapshots of the Windows Registry — one before and one after malware execution — and compares them to identify all changes. This reveals persistence mechanisms, configuration settings, and any other registry modifications the malware makes.

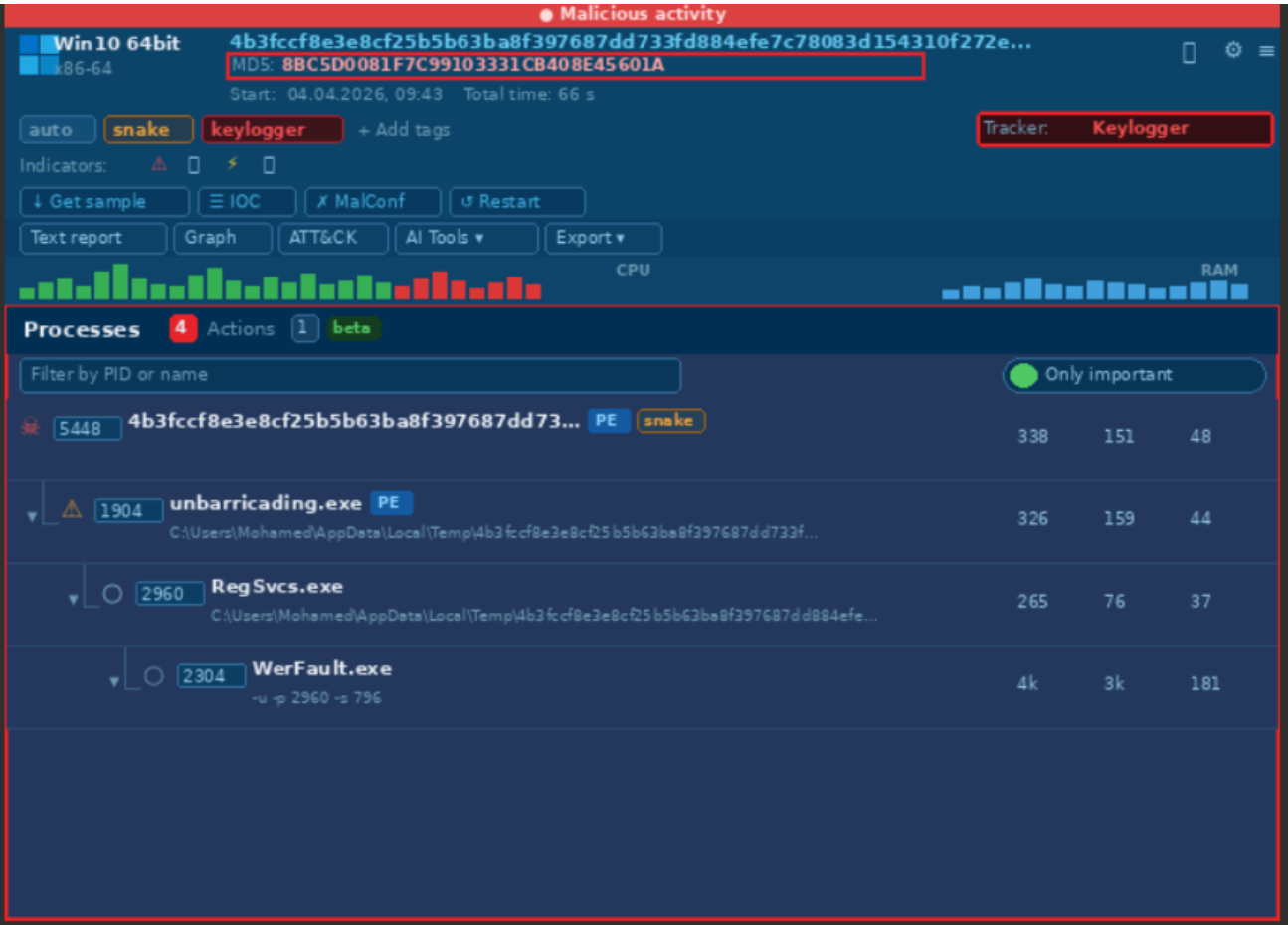
Registry changes identified after malware execution included:

- New keys added under HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run (autostart persistence)
- New values pointing to unbarricading.vbs in the Startup folder
- Modifications to security policy keys
- New service registration entries

## 6. HYBRID MALWARE ANALYSIS (ANY.RUN)

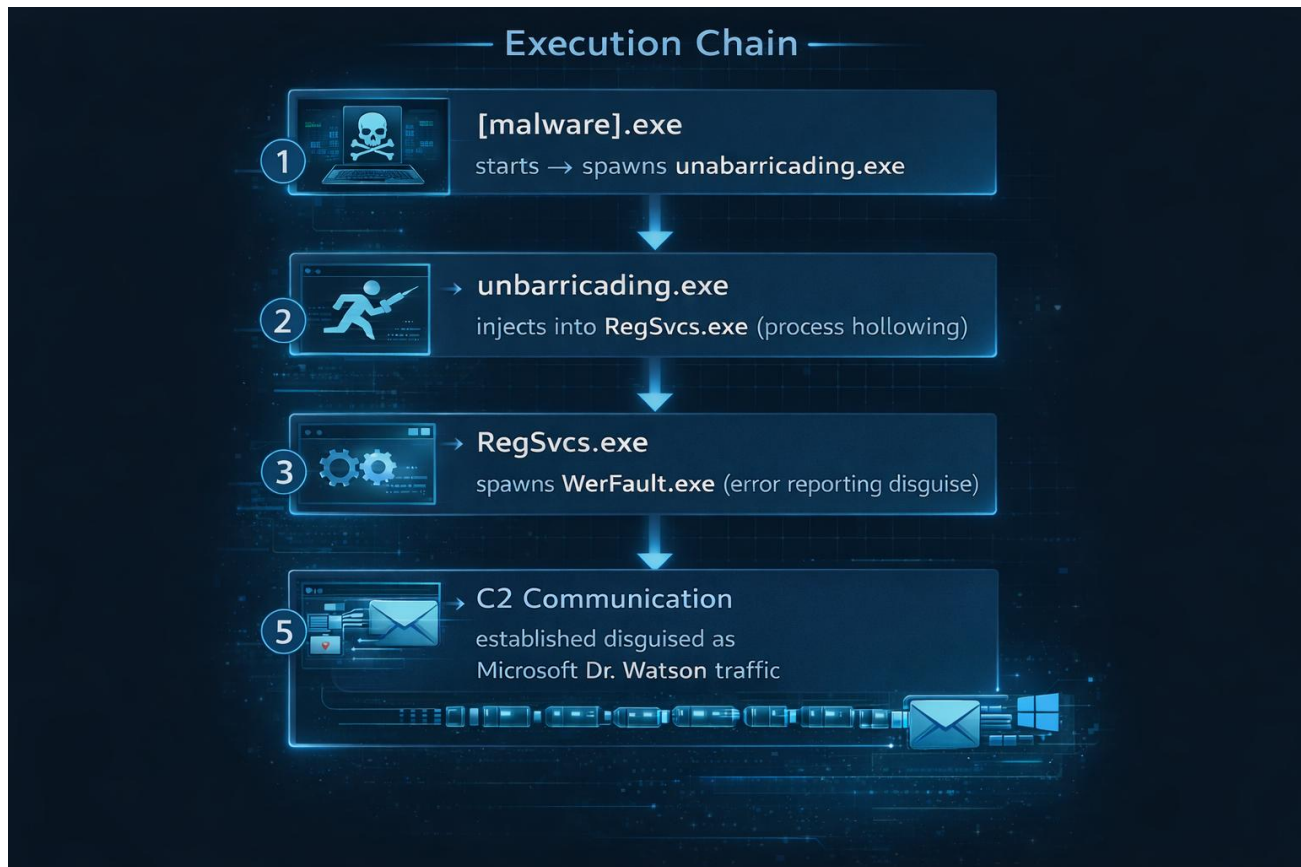
Hybrid analysis combines the depth of static analysis with the behavioral evidence of dynamic analysis. ANY.RUN, an interactive online malware sandbox, was used to perform a comprehensive automated analysis of the Snake Keylogger sample in a Windows 10 64-bit environment.





## 6.1 Process Graph

The ANY.RUN process graph revealed the complete execution chain of the Snake Keylogger:



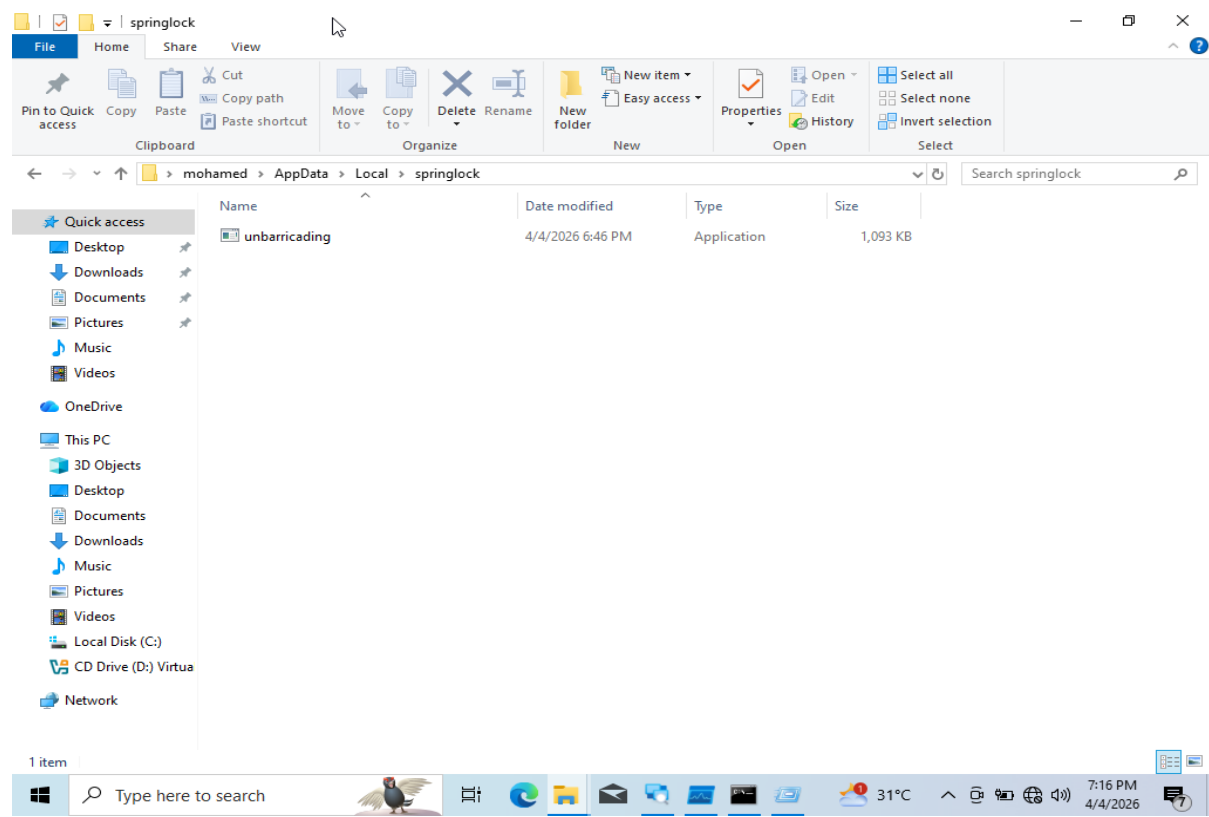
### EXECUTION CHAIN

1. [malware].exe starts → spawns unbarricading.exe 2. unbarricading.exe → injects into RegSvc.exe (process hollowing) 3. RegSvc.exe → spawns WerFault.exe (error reporting disguise) 4. WerFault.exe → sends C2 communication disguised as Microsoft Dr. Watson traffic

Analysis of each process:

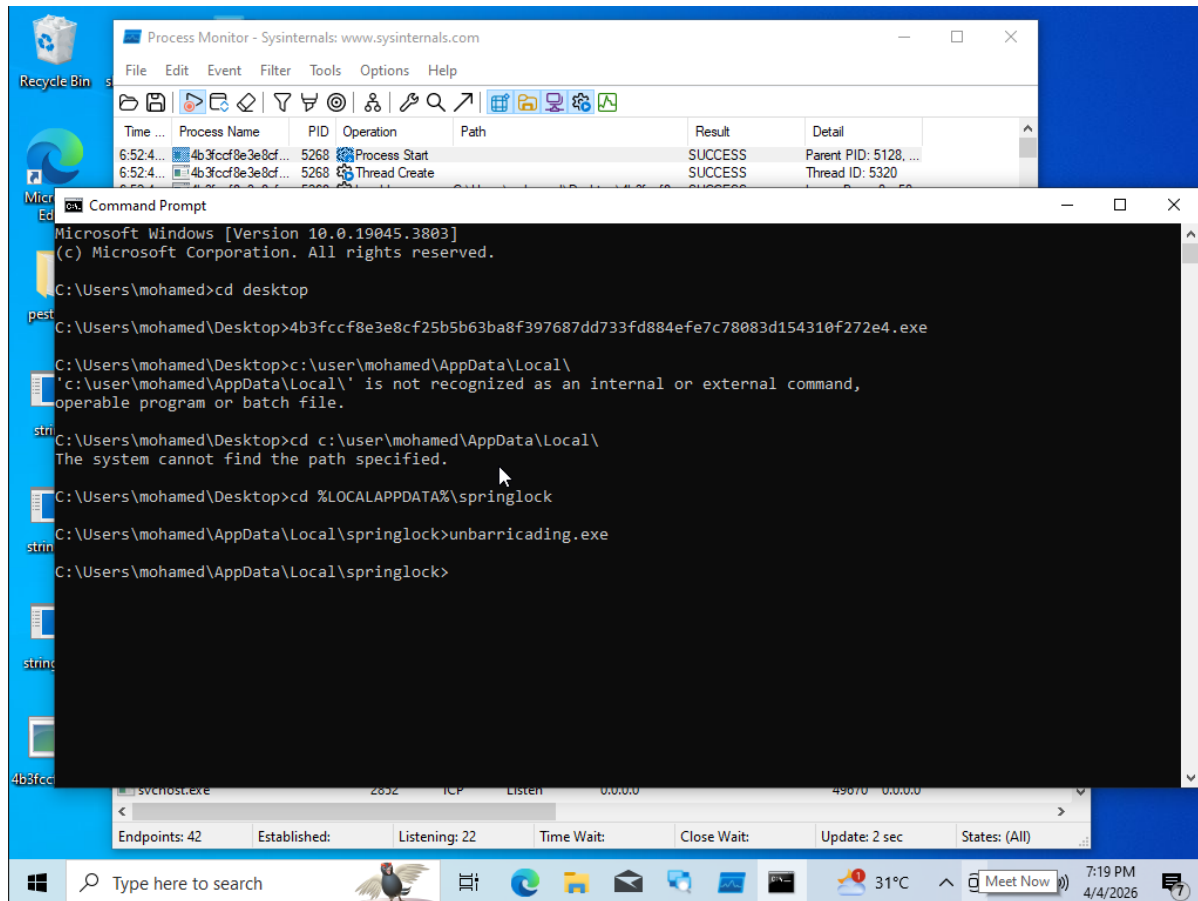
- unbarricading.exe (PID 1904) — Intermediate launcher dropped in AppData\Local\Temp. Creates persistence via Startup folder VBS script (unbarricading.vbs).
- RegSvc.exe (PID 2960) — Process hollowing target. Legitimate .NET process that hosts the injected Snake Keylogger payload. Performs all keylogging and C2 communication.
- WerFault.exe (PID 2304) — Windows Error Reporting process used as a decoy. Masks C2 HTTP traffic as legitimate Microsoft telemetry/crash reporting.

## 6.2 Malware Confirmation



| Property          | Value                                                     |
|-------------------|-----------------------------------------------------------|
| Verdict           | MALICIOUS ACTIVITY                                        |
| Malware Family    | Snake Keylogger (Keylogger Tracker)                       |
| MD5 Match         | 8BC5D0081F7C99103331CB408E45601A                          |
| OS                | Windows 10 Professional (build 19044, 64-bit)             |
| Analysis Tags     | auto, snake, keylogger                                    |
| MIME Type         | application/vnd.microsoft.portable-executable             |
| File Info         | PE32 executable (GUI) Intel 80386, MS Windows, 5 sections |
| Analysis Duration | 60 seconds                                                |

## 6.3 Behaviour Activities



MALICIOUS behaviors confirmed:

- **SNAKE has been found (auto-detected by ANY.RUN malware classification engine)**
- **Creates files in the Startup directory**  
(C:\Users\mohamed\AppData\Local\springlock\unbarricading.exe)

SUSPICIOUS behaviors observed:

- Executable content was dropped or overwritten by the primary process
- Malware starts itself from a different location (AppData\Local\Temp)
- Checks supported languages — targets specific language environments
- Reads mouse settings — captures mouse cursor position alongside keystrokes
- Reads machine GUID from registry — uniquely identifies victim machine for C2
- Creates files/folders in user directory for data storage
- Launching a file from the Startup directory (unbarricading.exe)
- Reads computer name from registry — collects system fingerprint

## 6.4 Files Activity

| Property               | Detail                                                                                         |
|------------------------|------------------------------------------------------------------------------------------------|
| Total Files Created    | 12 (1 executable, 9 suspicious, 2 text files)                                                  |
| Key Dropped File 1     | C:\Users\admin\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\unbarricading.vbs |
| Key Dropped File 2     | C:\ProgramData\Microsoft\Windows\WER\Temp\WERE00.tmp.dmp                                       |
| Dropped by             | WerFault.exe (PID 2304) — disguised as crash report                                            |
| C2 Files SHA-256       | 9966f4f9193403c9b1f92f8a16f6ad22770186e3e6ec104cda5523475d8063e0                               |
| Startup Script SHA-256 | 5dd26a5996a8fb517c6c00c424c165eec89e3d1e2cd1771dba8c8194ea89cb01                               |

## 6.5 Network Activity

| Network Property    | Value                                                                   |
|---------------------|-------------------------------------------------------------------------|
| HTTP(S) Requests    | 32                                                                      |
| TCP/UDP Connections | 27                                                                      |
| DNS Requests        | 21                                                                      |
| Detected Threats    | 1 (Unknown Traffic — fake Dr. Watson User-Agent)                        |
| Key C2 Domain       | login.live.com (Microsoft credentials phishing target)                  |
| Data Transmission   | POST requests to https://login.live.com/RST2.srf (1.24 KB sent)         |
| Masquerading        | WerFault.exe uses Microsoft Dr. Watson User-Agent (MSDW) for C2 traffic |

## 7. INDICATORS OF COMPROMISE (IOCs)

The following Indicators of Compromise were identified through static analysis, dynamic analysis, and hybrid sandbox analysis. These IOCs can be used to detect Snake Keylogger infections in enterprise environments using SIEM, EDR, and network monitoring tools.

### 7.1 File Hashes

| Hash Type                          | Value                                                                                       |
|------------------------------------|---------------------------------------------------------------------------------------------|
| MD5                                | 8bc5d0081f7c99103331cb408e45601a                                                            |
| SHA-1                              | 796ade52d238971ed98543c7b4670f5e746d22b1                                                    |
| SHA-256<br>(Primary)               | 4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4                            |
| SHA-256<br>(unbarricading.<br>exe) | 5dd26a5996a8fb517c6c00c424c165eec89e3d1e2cd1771dba8c8194ea89cb01                            |
| SHA-256<br>(WER dump)              | 9966f4f9193403c9b1f92f8a16f6ad22770186e3e6ec104cda5523475d8063e0                            |
| Imphash                            | 4b00809                                                                                     |
| SSDEEP                             | 49152:nP86ycz3PjS/c4x0exOwixol1pwpYBaBgx2H8eNkjpygO8eyGN71d0gFr0p08hq1nPOcb12H0q1roPXHq2HBA |

### 7.2 File System IOCs

| Location                   | File Path                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------|
| Startup Persistence Script | C:\Users\[user]\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\unbarricading.vbs         |
| Temp Executable            | C:\Users\[user]\AppData\Local\Temp\4b3fccf8e3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4.exe |
| WER Temp File              | C:\ProgramData\Microsoft\Windows\WER\Temp\WERE00.tmp.dmp                                                |
| Injected Process           | C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe                                               |
| Keylog Storage             | AppData\LocalLow\Microsoft\CryptnetUrlCache\Content\[hash]                                              |
| Dropped DLL Metadata       | C:\ProgramData\Microsoft\Windows\WER\Temp\WEREE0.tmp.WERInternalMetadata.xml                            |

## 7.3 Registry IOCs

| Registry Path        | Purpose                                                           |
|----------------------|-------------------------------------------------------------------|
| Persistence Key      | HKCU\Software\Microsoft\Windows\CurrentVersion\Run\[malware name] |
| System Fingerprint   | HKLM\SYSTEM\CurrentControlSet\Control\ComputerName\ComputerName   |
| Machine GUID         | HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid                  |
| Installed Programs   | HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\         |
| Language Query       | HKLM\SYSTEM\CurrentControlSet\Control\Nls\Language                |
| Video Card Detection | HKLM\SOFTWARE\Microsoft\WoW64\x86\VirtualBox (VM detection)       |

## 7.4 Network IOCs

| IOC Type             | Value                                                                                                                                      |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| C2 Target            | login.live.com (Microsoft account phishing)                                                                                                |
| HTTP User-Agent      | Microsoft-CryptoAPI/10.0 (masquerading as Windows crypto)                                                                                  |
| Fake User-Agent      | Microsoft Dr. Watson User-Agent (MSDW) via WerFault.exe                                                                                    |
| POST Endpoint        | https://login.live.com/RST2.srf (sends stolen credentials)                                                                                 |
| Certificate Check    | Queries CRL servers: <a href="https://crl.microsoft.com">crl.microsoft.com</a> , <a href="https://ocsp.digicert.com">ocsp.digicert.com</a> |
| Telemetry Masquerade | <a href="https://watson.events.data.microsoft.com/Telemetry.Request">https://watson.events.data.microsoft.com/Telemetry.Request</a>        |
| DNS Queries          | 21 DNS lookups to resolve C2 and masquerade domains                                                                                        |

## 7.5 Process & Behavioral IOCs

| Behavior         | Description                                                        |
|------------------|--------------------------------------------------------------------|
| Injected Process | RegSvc.exe spawned with malware command line argument              |
| Decoy Process    | WerFault.exe -u -p [PID] -s 796 (fake crash report)                |
| VBS Execution    | wscript.exe executing unbarricading.vbs from Startup folder        |
| Process Ancestry | malware.exe → unbarricading.exe → RegSvc.exe → WerFault.exe        |
| Memory Pattern   | Process hollowing: RegSvc.exe memory replaced with malware payload |
| Anti-VM Check    | Registry queries for VirtualBox/VMware detection signatures        |

## 8. YARA DETECTION RULE

The following YARA rule was developed based on unique characteristics identified during static and dynamic analysis of the Snake Keylogger sample. This rule can be deployed in endpoint security solutions, SIEM platforms, and file scanning pipelines to detect new variants of this malware family.

```
rule Snake_Keylogger_AutoIt_Variant {
  meta:
    author      = "Laiba Malik — SOC Analyst"
    description = "Detects Snake Keylogger compiled with AutoIt3"
    sha256      = "4b3fccf8c3e8cf25b5b63ba8f397687dd733fd884efe7c78083d154310f272e4"
    malware_family = "SnakeKeylogger"
    severity    = "HIGH"
    date       = "2026-03-13"
    reference   = "bazaar.abuse.ch/sample/4b3fccf..."

  strings:
    // AutoIt3 builder signature
    $autoit_magic = { 41 75 74 6F 49 74 } // "AutoIt"
    $autoit_ver   = "AutoIt3 [v3.3.3]" ascii nocase
    // Snake Keylogger persistence artifacts
    $startup_vbs  = "unbarricading.vbs" ascii nocase wide
    $startup_exe  = "unbarricading.exe" ascii nocase wide
    // Process hollowing target
    $hollow_target = "RegSvcs.exe" ascii nocase wide
    // Network masquerading strings
    $werfault_ua   = "Microsoft Dr. Watson" ascii nocase
    $c2_login      = "login.live.com" ascii nocase wide
    $http_send     = "HttpSendRequestW" ascii
    $http_open     = "HttpOpenRequestW" ascii
    // Keylogging function imports
    $key_api       = "GetAsyncKeyState" ascii
    $clip_api      = "GetClipboardData" ascii
    // Registry persistence import
    $reg_persist   = "RegSetValueExW" ascii
    $reg_path      = "Software\\Microsoft\\Windows\\CurrentVersion\\Run" ascii nocase wide
    // Anti-analysis techniques
    $anti_vm1      = "VBOX" wide ascii nocase
    $anti_vm2      = "VMware" wide ascii nocase
    $tick_count    = "GetTickCount" ascii

  condition:
    uint16(0) == 0x5A4D and           // Valid PE/MZ header
    filesize < 5MB and                // File size constraint
    ($autoit_magic or $autoit_ver) and // AutoIt builder
    ($startup_vbs or $startup_exe) and // Persistence
    $hollow_target and                // Process hollowing
    ($key_api and $reg_persist) and    // Keylogging + persist
    ($http_send or $http_open or $c2_login) // C2 communication
}
```

## 9. MITRE ATT&CK FRAMEWORK MAPPING

The behaviors observed during analysis were mapped to the MITRE ATT&CK framework. This provides standardized terminology for threat reporting and helps security teams implement targeted detection rules.

| Technique ID & Name                            | How Snake Keylogger Uses It                                             |
|------------------------------------------------|-------------------------------------------------------------------------|
| T1059.005 — Command & Scripting: Visual Basic  | Uses VBScript (unbarricading.vbs) for execution and persistence         |
| T1547.001 — Registry Run Keys / Startup Folder | Drops unbarricading.vbs in Windows Startup folder for auto-execution    |
| T1055.012 — Process Hollowing                  | Injects malicious code into RegSvcs.exe using process hollowing         |
| T1056.001 — Keylogging                         | Core capability — uses GetAsyncKeyState to capture every keystroke      |
| T1012 — Query Registry                         | Extensively queries registry for system fingerprinting and VM detection |
| T1082 — System Information Discovery           | Reads computer name, GUID, language settings, hardware info             |
| T1497.001 — System Checks (Virtualization)     | Queries VirtualBox/VMware registry keys to detect analysis environment  |
| T1071.001 — Web Protocols (HTTP/S)             | Uses HTTP/HTTPS to communicate with C2 infrastructure                   |
| T1036.005 — Match Legitimate Name or Location  | Uses WerFault.exe (Windows Error Reporting) to disguise C2 traffic      |
| T1041 — Exfiltration Over C2 Channel           | Sends stolen keystrokes and credentials via HTTP POST to C2 server      |
| T1027 — Obfuscated Files or Information        | High entropy .rsrc section (7.79/8.0) — encrypted payload               |
| T1140 — Deobfuscate/Decode Files               | Decrypts embedded AutoIt script at runtime for execution                |

## 10. INCIDENT RESPONSE PLAN

The following incident response plan was developed based on the observed behavior of Snake Keylogger. This plan follows the standard NIST SP 800-61 incident response lifecycle: Detection, Containment, Eradication, and Recovery.

### Phase 1: Detection — Identifying the Infection

Immediate detection indicators to look for when Snake Keylogger is suspected:

- Search for process unbarricading.exe running from AppData\Local\Temp
- Look for RegSvcs.exe with an unusual command line argument (pointing to malware path)
- Monitor for creation of unbarricading.vbs in the Windows Startup folder
- Check for file hash matches against known Snake Keylogger SHA-256 hashes in EDR/AV logs
- Alert on GetAsyncKeyState API calls from non-keyboard-software processes
- Monitor DNS queries for unusual patterns — 21+ DNS lookups in rapid succession
- Flag HTTP POST requests from WerFault.exe to login.live.com or unknown IPs

### Phase 2: Containment — Stopping the Spread

Immediate actions to contain the infection and prevent further data loss:

- **IMMEDIATELY disconnect the infected machine from the network (pull Ethernet, disable Wi-Fi)**
- Block outbound connections to known C2 domains at the perimeter firewall: login.live.com (for attacker-controlled variants), watson.events.data.microsoft.com (if abnormal)
- Disable the Windows Startup folder VBS script: del "C:\Users\[user]\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\unbarricading.vbs"
- Kill suspicious processes in this order: WerFault.exe (decoy C2 relay) → RegSvcs.exe (injected payload) → unbarricading.exe (launcher)
- Take a memory dump of the infected machine using Volatility for forensic preservation before killing processes

## Phase 3: Eradication — Removing the Malware

Complete removal of all malware artifacts from the system:

- Delete all identified malicious files from AppData\Local\Temp, AppData\LocalLow, and WER\Temp directories
- Remove registry persistence keys under HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- Scan with multiple AV engines (Microsoft Defender, Malwarebytes, ESET Online Scanner) to catch any remaining components
- Re-image the system if sensitive credentials (banking, email, corporate VPN) may have been captured — clean reinstall is the safest option
- Audit all AutoIt-compiled executables on the network — Snake Keylogger frequently uses AutoIt and any similar unknowns should be sandboxed

## Phase 4: Recovery — Restoring Operations

Actions required to safely restore the system and prevent re-infection:

- **CHANGE ALL PASSWORDS from a clean, uninfected device — email, banking, corporate VPN, and any accounts accessed on the infected machine during the infection window**
- Enable Multi-Factor Authentication (MFA) on all accounts that may have been compromised
- Restore the system from a clean snapshot/backup taken before the infection date
- Apply all outstanding Windows security updates and patches
- Deploy the YARA rule from Section 8 to your EDR/SIEM for ongoing detection
- Review network logs for evidence of successful data exfiltration — check for POST requests to C2 servers during the infection window
- Brief the affected user on phishing recognition — Snake Keylogger commonly arrives via malicious email attachments

## Phase 5: Lessons Learned

| Control                  | Recommendation                                                                    |
|--------------------------|-----------------------------------------------------------------------------------|
| Application Whitelisting | Block execution of AutoIt-compiled executables from AppData and Temp directories  |
| Script Block Logging     | Enable PowerShell and VBScript logging to catch malicious script execution        |
| LOLBAS Monitoring        | Alert on RegSvcs.exe executing with unusual parent processes or command lines     |
| Email Filtering          | Block .exe, .vbs, .au3 file attachments in email gateways                         |
| EDR Coverage             | Ensure EDR with memory scanning is deployed on all endpoints                      |
| User Training            | Conduct phishing simulation training focusing on executable disguised attachments |

# 11. CONCLUSION & RECOMMENDATIONS

This laboratory exercise successfully demonstrated a complete malware analysis workflow against a real-world Snake Keylogger sample. The analysis confirmed that the malware is a sophisticated information-stealing trojan with the following key capabilities:

- **KEYLOGGING:** Continuous keystroke capture using GetAsyncKeyState (called 263 times in code)
- **PROCESS HOLLOWING:** Injection into legitimate RegSvcs.exe to evade detection
- **PERSISTENCE:** VBS startup script ensures the malware survives system reboots
- **C2 COMMUNICATION:** HTTP exfiltration of stolen credentials to attacker servers
- **EVASION:** VM detection, process masquerading via WerFault.exe, encrypted .rsrc section

The hybrid analysis approach — combining manual static analysis (pecheck, Exeinfo PE, Ghidra), dynamic behavioral monitoring (Procmon, Process Explorer), and automated sandbox analysis (ANY.RUN) — provided comprehensive visibility into the malware's full behavior chain. Each method contributed unique insights that no single approach could provide alone.

## ANALYST ASSESSMENT

Snake Keylogger represents a medium-to-high threat level for enterprise environments. Its use of legitimate Windows processes for hollowing (RegSvcs.exe), AutoIt compilation for AV evasion, and masquerading of C2 traffic as Microsoft telemetry make it difficult to detect with signature-based tools alone. Behavioral detection rules based on the MITRE ATT&CK mappings and YARA rule in this report provide the most effective detection strategy.

## 12. REFERENCES

1. MalwareBazaar — Snake Keylogger sample repository: <https://bazaar.abuse.ch/browse/>
2. ANY.RUN Interactive Sandbox analysis: <https://app.any.run/tasks/c12ac88e-ad31-4899-af90-6dedc158464f>
3. REMnux Linux distribution for malware analysis: <https://remnux.org/>
4. Ghidra reverse engineering framework: <https://github.com/NationalSecurityAgency/ghidra/releases>
5. Sysinternals Process Monitor & Process Explorer: <https://docs.microsoft.com/sysinternals>
6. MITRE ATT&CK Framework: <https://attack.mitre.org/>
7. INetSim network simulator: <https://www.inetsim.org/>
8. Java OpenJDK (Temurin): <https://adoptium.net/temurin/releases>
9. Exeinfo PE tool: <https://github.com/ExeinfoASL/Exeinfo>
10. ChatGPT AI assistant (analysis guidance): <https://chatgpt.com/share/69d132ae-5a98-83a6-bc64-854dcb295750>
11. Laiba Malik LinkedIn: <https://linkedin.com/in/laiba-malik-soc>

— END OF REPORT —